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Symptoms of Anxiety in Adults with ADHD: A Systematic Review and Meta-Analysis of Case-Control Studies

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Abstract

Objective: Research suggests there is a higher prevalence of anxiety in individuals with ADHD compared to those without ADHD. However, estimated prevalence rates vary greatly due to differences in methodologies used across studies. This study is the first meta-analysis investigating differences in anxiety measure scores between adults with and without ADHD, and possible moderators of this effect.

Method: Our analysis included 58 studies that compared anxiety scores in adults with/without ADHD ($N = 18{,}821$; $k = 112$).

Results: The average effect size (Hedges's *g*) of anxiety scores in those with ADHD relative to controls was 0.77 ($SE = 0.066$); a medium effect. The only significant moderator was the type of comparison group used, with comparisons to non-clinical control groups ($g = 0.954$, $SE = 0.08$) yielding significantly larger effects than comparisons to clinical control groups ($g = 0.437$, $SE = 0.09$). None of the other moderator variables examined (Method of characterising ADHD, Anxiety Measure Type, Anxiety Measure Focus, Age or Gender) moderated the effect. There was some evidence of publication bias, therefore results should be interpreted with this in mind.

Conclusion: Overall, this study indicates that ADHD is associated with higher levels of anxiety symptomology. This has important implications in the diagnosis and treatment of individuals presenting with ADHD or anxiety, including the consideration and screening of each of these conditions as potentially co-occurring with the other. Adaptions made to clinical practice in line with this would better support this population, improve symptom management and overall quality of life. Suggestions for future research are discussed (*J. of Att. Dis. 2026; 30(10) 1305-1337*).

Keywords

attention deficit hyperactivity disorder (ADHD), adults with ADHD, anxiety symptomology, meta-analysis

Introduction

Attention deficit hyperactivity disorder (ADHD) is a neurodevelopmental condition that impacts individuals' daily functioning with symptoms of inattentiveness and/or hyperactivity and impulsivity (American Psychological Association [APA], 2013). Individuals with ADHD can experience forgetfulness, difficulties with planning and organisation, restlessness and mood instability (Ambrosini et al., 2013). ADHD can affect an individual's relationships, education and work (Michielsen et al., 2018) as well as general adaptive functioning and quality of life (Meaux et al., 2009).

ADHD has historically been viewed as a childhood disorder that improves with age; however, recent research has shown that ADHD continues into adulthood (Johnson et al., 2021; Michielsen et al., 2018), impacting around 2% to 5% of adults worldwide (Ayano et al., 2023). Despite its prevalence adult ADHD is poorly understood. However, meta-analyses have shown that the prevalence rate of ADHD is higher within adult populations that have concurrent mental health diagnoses, although there is substantial variation in this prevalence estimate varying from 6.9% to 38.75% depending on the ADHD assessment used and sampling method (Gerhand & Saville, 2022). Notably, Adamis et al. (2022) is the only meta-analysis to report a pooled prevalence rate, estimating that 26.7% of psychiatric outpatients have ADHD. This highlights the high co-occurrence of mental health difficulties in ADHD, yet there remains a lack of evidence on whether the severity of mental health symptoms differ in this population compared to those without ADHD, and whether this is true of specific mental health diagnoses, for example, anxiety.

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ADHD and Co-occurring Conditions

ADHD has been found to co-occur with several mental health conditions, neurodevelopmental conditions, and substance misuse, within both cross-sectional (e.g. Halmøy et al., 2009) and longitudinal studies (e.g. Biederman et al., 2006). Indeed, 75% of adults with ADHD are thought to have at least one other psychiatric diagnosis (Kooij et al., 2010).

Systematic reviews have found the most common co-occurring conditions in adults with ADHD are substance use disorder (2.3%–82.9%), mood disorders (4.45%–80.8%), anxiety disorders (3.9%–84%) and personality disorders (0.31%–65.95%) (Choi et al., 2022; Katzman et al., 2017). Choi et al. (2022) concluded that prevalence rates of these conditions are higher in samples with ADHD compared to samples without ADHD, irrespective of the presence of any other prior mental health diagnoses. However, this research (Choi et al., 2022) did not include any statistical analysis to quantify group differences or explore possible moderators. The wide range of prevalence estimates of co-occurring mental health conditions in ADHD is likely due to methodological differences across studies, such as whether the sample was from the general public or a clinical population, and how ADHD and comorbid mental health conditions were each diagnosed (Choi et al., 2022; Katzman et al., 2017). Prevalence estimates are less varied when study methods are controlled (Polanczyk, et al., 2014).

The rate and severity of current and lifetime mental health conditions can vary based on whether someone was diagnosed with ADHD in childhood or adulthood (Able et al., 2007; Halmøy et al., 2009; Quintero et al., 2019). ADHD has been seen to affect the clinical course of mental health conditions (Biederman et al., 2006), including increased rates of episodes (Rydén et al., 2009) and functional impairment (Asherson et al., 2012). Co-occurring conditions have also been seen to increase with age, particularly with life transitions such as when individuals leave their family home (Barkley et al., 2002). Moreover, ADHD populations have been found to have lower quality of life, often in relation to severity of ADHD symptomology, social functioning and emotional control (Agarwal et al., 2012). Increased ADHD symptomology also increases the likelihood of self-harm and suicidal behaviours (Austgulen et al., 2023), potentially in relation to impulsivity being a core symptom in ADHD (APA, 2013). The presence of co-occurring mental health disorders such as affective disorders, anxiety disorders and substance misuse, have been found to mediate the relationship between ADHD and suicidality (Balazs & Kereszteny, 2017) likely in relation to difficulty coping with the impact of these on quality of life (Furczyk & Thome, 2014). Given the significant role of anxiety disorders in mediating the relationship between ADHD and quality of life, as well as their high co-occurrence, further exploration of this comorbidity is essential to understanding its impact.

ADHD and Anxiety

ADHD and anxiety disorders frequently co-occur (Biederman et al., 2006; Katzman et al., 2017; Kessler et al., 2006; Lynch & Davison, 2022; O'Rourke et al., 2020), particularly social anxiety, specific phobias and post-traumatic stress disorder (Kessler et al., 2006), with individuals meeting diagnostic criteria for both conditions, even when potential overlapping symptoms like physical restlessness are excluded (Milberger et al., 1995). The prevalence of co-occurring anxiety disorders has been found to range from 4.3% to 47.1% in individuals with ADHD sampled from the general population and 3.9% to 84% when the ADHD sample is taken from a psychiatric population (Choi et al., 2022). Hartman et al. (2023) found similar variability in prevalence rates (2.2%–44.3%) in their systematic review comparing anxiety in ADHD adults and controls which resulted in a pooled odds ratio of 5.0, yet it is unclear the extent to which the symptomology and severity of anxiety is the same or different between these groups.

Research examining the impact of this co-occurrence has found that anxiety can change the presentation, course and prognosis of ADHD across the lifespan (D'Agati et al., 2019; Schatz & Rostain, 2006). Anxiety has been found to worsen cognitive deficits seen in ADHD, particularly working memory (Roth et al., 2004). Furthermore, anxiety sometimes interferes with impulsivity and inhibitory control deficits in ADHD (Schatz & Rostain, 2006). There are mixed findings as to whether this improves performance or not (Brown, 2000; Oosterlaan & Sergeant, 1998), but anxiety has been found to negatively impact the self-esteem of individuals with ADHD (Brown, 2000). Research has also indicated challenges in treatment for individuals with co-occurring ADHD and anxiety (D'Agati et al., 2019; Koyuncu et al., 2022). For example, responses to methylphenidate treatment for ADHD are varied in individuals with co-occurring anxiety (Koyuncu et al., 2022) and these individuals may benefit less from cognitive behavioural treatment alone, needing supplementary medication (D'Agati et al., 2019). It is therefore important for assessments to carefully consider the developmental course of the conditions to reduce the likelihood of misdiagnosis and ensure effective treatment (D'Agati et al., 2019).

Several theories aim to explain the increased prevalence of anxiety disorders in ADHD (D'Agati et al., 2019; Jarrett & Ollendick, 2008; Schatz & Rostain, 2006). Research has highlighted a potential genetic underpinning to comorbidities, utilising latent class analysis to analyse patterns to indicate subgroups (Neuman et al., 2001) or highlighting a potential underlying risk factor due to high rates of familial association (Braaten et al., 2003). However, this means it can be hard to differentiate nature and nurture in the interplay of ADHD and anxiety. The Cognitive Processing Model (Schachar et al., 1995) proposes that symptoms of ADHD lead to difficulties conforming with norms and

meeting demands thus resulting in anxiety. The Executive Inhibition (Barkley, 1997) and Motivational models (Quay, 1998) similarly consider the impact of socially inappropriate behaviour, but add the potential of executive functioning difficulties resulting in undefined anxiety.

Other models have considered the impact of poor regulation (Brown, 2000) and coordination (Posner & Peterson, 1990) in attentional systems in ADHD leading to increased focus on threatening stimuli and decreased regulation of emotion, that is, anxiety. Levy (2004) created a hybrid theory to address the contradiction that individuals with ADHD can be both impulsive and daring, yet experience high anxiety (Schatz & Rostain, 2006). This integrates prior theories and considers both cortical and subcortical malfunction in ADHD leading to difficulties in regulating attention and focus on anxiety provoking stimuli and poor regulation of anxiety states (Levy, 2004). These theories offer insights into the link between ADHD and anxiety, suggesting how genetic, neurobiological and environmental factors may each contribute to anxiety in individuals with ADHD.

Rationale for the Present Study

As discussed, there is high variability in the estimated prevalence of co-occurring ADHD and anxiety which could be due to sampling and diagnostic differences across previous studies (Davies et al., 2022). Behavioural studies converge on the idea that individuals with ADHD experience a greater level of anxiety than controls (Agarwal et al., 2012; Nelson & Liebel, 2018). However, overlapping symptoms and subjective experiences of these conditions can make their differentiation difficult in self-identification (Koyuncu et al., 2022). Previous research has shown a discrepancy between ADHD participants who self-reported anxiety compared to those who met the clinical criteria for a lifetime anxiety disorder (Halmøy et al., 2009). Therefore, to further understand the relationship between ADHD and anxiety this study only includes scores on questionnaire measures of anxiety, and not self-reported anxiety, as these are generally deemed more objective than relying on participant report of a diagnosis (Hamilton et al., 2017).

Meta-analytic data is needed to quantify the levels of anxiety in individuals with ADHD compared to non-ADHD control groups beyond binary classifications of anxiety being present or not. This will enable the estimation of differences in anxiety symptomology between adults with and without an ADHD diagnosis and aid better understanding of how these conditions co-occur (Gorlin et al., 2016).

Much of the previous research on ADHD has focused on children (Ginapp et al., 2022), but there is a growing interest and focus on ADHD in adulthood (Ramos-Quiroga et al., 2013). This study focuses on the relationship between ADHD and anxiety in adulthood for three main reasons. Firstly, the presence of co-occurring conditions has been seen to affect the diagnosis and treatment of ADHD in adulthood due to a lack of appropriate resources (Ramos-Quiroga et al., 2013). Secondly, co-occurring anxiety has been seen to increase with age alongside changes in overt symptom presentation (Koyuncu et al., 2022). Finally, research with children can often rely on parental ratings of child anxiety which can be inaccurate (Schatz & Rostain, 2006).

Improving understanding of the relationship between ADHD and anxiety could support healthcare professionals in their understanding of ADHD presentations (Ginapp et al., 2022). Understanding disorder-related outcomes would have implications for recognition, assessment and treatment (Jarrett & Ollendick, 2008), which can be important in suicide prevention (Balazs & Kereszteny, 2017) and improving quality of life for adults with ADHD. This in turn may reduce the impact of these conditions (Davies et al., 2022), and potentially change psychiatric trajectories (Katzman et al., 2017).

Moderators. As methodological differences are common in studies on this topic, several moderators were considered. Firstly, prevalence rates of both ADHD and co-occurring anxiety seem to be affected by variability in diagnostic tools and methodologies (Choi et al., 2022) and inconsistencies in how ADHD is defined (Koyuncu et al., 2022). Therefore, we examined the impact of how ADHD was diagnosed and differentiated from non-ADHD groups as a moderator, for example, self-report compared to clinical diagnosis.

We also considered the nature of the control group, that is, whether controls were recruited from the general or clinical population.

Furthermore, due to heterogeneity in how anxiety is measured within studies, it was important to examine whether this moderated effects, including comparisons between the type of anxiety measured (e.g. general and specific anxieties), and whether measured focused on state or trait anxiety. Variability in anxiety measurement can influence research findings, potentially affecting the strength or direction of relationships between anxiety and other variables (Davies et al., 2022).

There is ongoing debate regarding whether there is a gender difference in the prevalence of ADHD with ADHD traditionally being associated with boys (Adamis et al., 2022). However, there is some evidence that females are more impaired on ADHD scales (Robison et al., 2008) and have more co-occurring psychiatric diagnoses (Ginapp et al., 2022). Indeed, prevalence rates of ADHD are likely similar among genders and females are potentially underdiagnosed, particularly in childhood, due to differences in symptom presentation, particularly a higher level of emotional symptoms (Adamis et al., 2022; Robison et al., 2008). Research has also consistently found higher prevalence rates of anxiety in females than males (Bandelow &

Michaelis, 2015). Therefore, gender was considered as a moderator in this analysis.

Age related changes in the association between ADHD symptomology and anxiety have been reported (D'Agati et al., 2019), thus age was also considered as a moderator. Finally, the publication status of the research was included as a moderator to examine the risk of publication bias.

Research Questions

This study addresses two main research questions. Firstly, is there a difference in anxiety measure scores between adults with and without ADHD? Secondly, is this difference moderated by participant or methodological variables? Within this, we specifically considered the effects of (a) method of categorising ADHD; (b) control group type; (c) anxiety measure type; (d) anxiety measure focus; (e) gender; (f) age; and (g) publication status.

Method

Database Search and Inclusion Criteria

The systematic review and meta-analysis were conducted in line with Preferred Reporting Items for Systematic reviews and Meta-Analyses (PRISMA) guidelines (Page et al., 2021). Searches were conducted in PsychINFO, MEDLINE, ERIC, Scopus and ProQuest databases for publications and dissertations. Searches included terms related to ADHD ("Attention Deficit Disorder", "Attention Deficit Hyperactivity Disorder", "ADHD", "Hyperkinetic Syndrome", "Minimal Brain Dysfunction) and for Anxiety ("Anxiety", "Phobia", "Sociophobia", "Panic", "Agoraphobia", "Claustrophobia", "GAD", "Hypochondria") within the title or abstract (see Appendix A for the full search strings used). The operationalisation of anxiety, and subsequent search terms were decided based on the categories of anxiety disorder featured within the DSM V (APA, 2013). A request for unpublished data was also distributed by the research team to their personal networks, as well as to COGDEVSOC and EUNETHYDIS listservs for distribution to their mailing lists (see Appendix B for the call for data).

The study inclusion criteria were that the study needed to be (a) written in English or have a suitable translation available, (b) contain new data, and (c) be completed/published from 1994 to June 2024. The start date was selected based on the publication of the fourth volume of the Diagnostic and Statistical Manual of Mental Disorders (DSM-IV; APA, 1994), which demonstrated a shift in the definition and understanding of ADHD and the inclusion of the three subtypes seen today (Epstein & Loren, 2013). The end date was the date the literature search was completed.

Furthermore, studies needed to (d) include participants aged 18 and over and (e) utilise a between-group design with one group categorised as having ADHD through either clinical diagnosis, self-report of prior diagnosis, or questionnaire with cut-off, and another control group without ADHD. Finally, (f) the study needed to include at least one measure of anxiety.

Data Screening and Extraction

Results from the database searches were uploaded to Rayyan software, and duplicates were removed. A two-stage screening process was undertaken as shown in Figure 1. In stage one, joint title and abstract screening was undertaken by the first author (89% articles) and two second coders (11% articles each). Half of all papers screened were double screened, that is, reviewed for a second time by one of the other coders resulting in a 97% inter-rater agreement. Any papers with disagreements in ratings were discussed with reference to the inclusion criteria and resolved.

100 articles progressed to stage two for full text screening. This was completed by the first author, with 15% of these screened a second time by a second coder (research student), giving 86.7% inter-rater agreement.

The first author independently collected and collated data from the included studies, including the mean and standard deviation on the anxiety measure for the ADHD group and any non-ADHD groups. Authors were contacted if the paper did not provide sufficient data to calculate effect sizes. Of these, 11 did not respond so were excluded from the meta-analysis, resulting in 58 studies included.

Each measure of ADHD was reviewed ensuring that the directionality of each measure was the same with a higher score meaning higher anxiety.

Moderator Variables

The operational definitions for moderator variables are presented below.

Method of Categorising ADHD. ADHD categorisation was coded categorically as a binary variable. Studies were categorised as "clinical" if ADHD group was determined using a researcher interview, with or without a questionnaire (Royal College of Psychiatrists, 2017), or participants had a confirmed diagnosis from a clinic. Alternately, the "self-report" category was used for studies who grouped participants based on participant-report of ADHD or questionnaire only.

Control Group Type. For this meta-analysis we compared ADHD groups with a control group who did not have ADHD. Differences in the type of control group were examined as a moderator.

The type of control group was a binary variable, coded categorically for each study. The label "non-clinical control" was used if the group was from the public, "clinical control" was used for participants recruited from

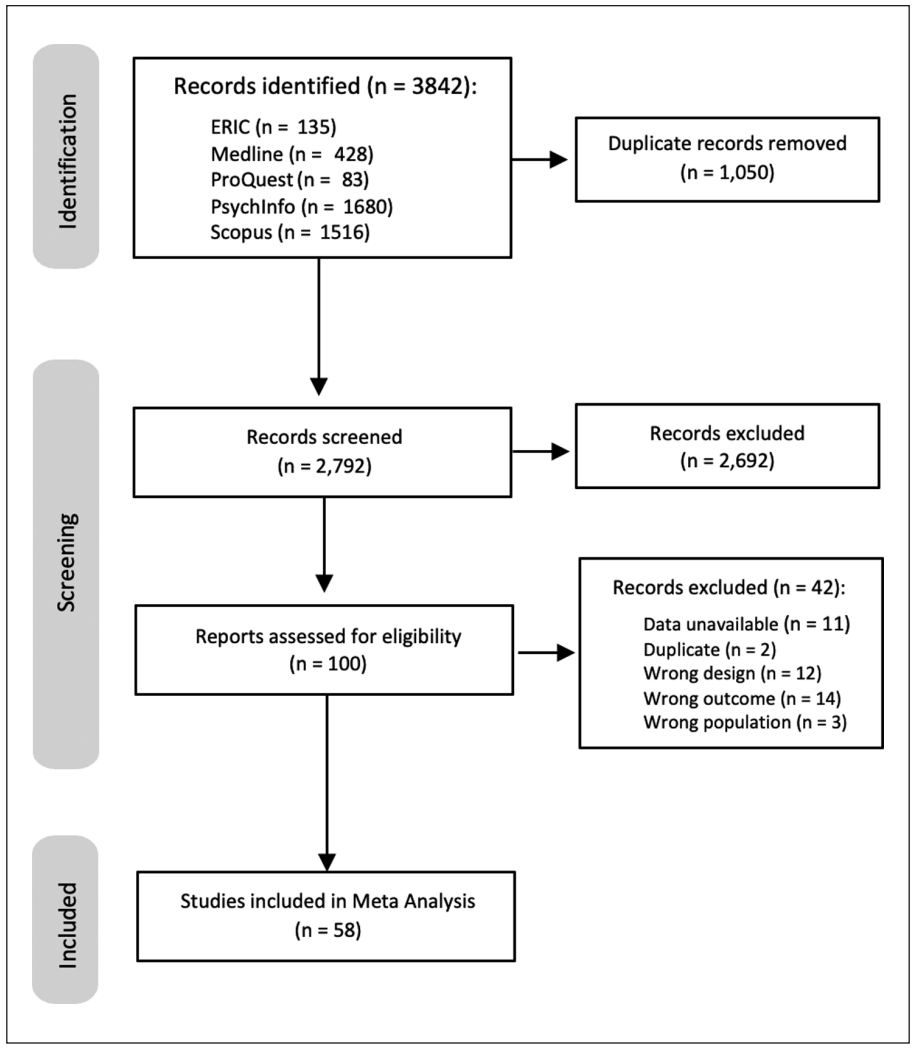


Figure 1. PRISMA diagram outlining search strategy and reasons for excluded studies.

psychiatric or hospital settings or where participants were determined to have a specific clinical diagnosis (e.g. bipolar disorder/ social anxiety disorder).

Anxiety Measure Type. For each study, two specific characteristics of the anxiety measure were considered. Firstly, the type of anxiety measured was coded categorically as either "general" if the measure assessed broad anxiety symptoms (e.g. Beck Anxiety inventory; Beck et al., 1988), or "specific" if the measure focused on a particular type of anxiety only, for example, Liebowitz Social Anxiety Scale; LSAS (Liebowitz, 1987).

Anxiety Measure Focus. Secondly, the form of anxiety was considered. The anxiety measure used was coded categorically as either "state" for anxiety measures that focused on how a person was feeling at that specific moment, or "trait" for measures assessing more stable individual tendencies (Leal et al., 2017). Within this research, we also included the category of "prolonged-state" (Hood et al., 2015) where the measure considers symptoms of anxiety over an extended period, that is, 2 weeks to 1 month.

Gender. The gender ratio of participants in the ADHD group was coded continuously as the percentage of females in the ADHD group compared to the total number of participants in the group.

Age. The age of ADHD group participants was coded continuously based on mean age in years. As this research was interested in whether the age of the ADHD participants impacted on relations with anxiety symptoms, due to evidence indicating changes in symptomology with age (D'Agati et al., 2019), the age of the control group was not used as a moderator.

Type of Publication. For each study, the publication status was coded categorically as "peer reviewed publication" if

the data was obtained from an article within a journal or "dissertation" if the data was from unpublished university theses and dissertations.

Analytical Approach

All analysis was completed in RStudio (Version 2024.12.1+563; see Appendix B for output). The Rcode for this analysis will be openly available on our OSF page upon publication (https://osf.io/rh6du/). The mean and standard deviation of the anxiety measure for each ADHD and non-ADHD group were used to calculate the Cohen's *d* effect sizes (Borenstein et al., 2009). The Cohen's *d* effect sizes were then converted to Hedges' *g* values by multiplying them by a correction factor based on the pooled sample size (Turner & Bernard, 2006). The standardised mean difference statistic of Hedges' *g* was used to estimate the mean difference between ADHD group and non-ADHD group scores as this corrects for small sample bias (Hedges, 1981). Hedges' *g* values are generally interpreted the same as the guidelines for Cohen's *d* (Brydges, 2019). They are interpreted as 0.2 being a small effect, 0.5 a medium, and 0.8 a large effect (Cohen, 1988, 1992).

Meta-analyses and metaregressions were completed using the robumeta package (Fisher & Tipton, 2015) in RStudio adapted from code used in Hawes et al. (2022). Specifically, robust variance estimation (RVE) was used due to some studies contributing multiple effect sizes to the meta-analysis as they included more than one anxiety measure comparison between the ADHD and control groups. RVE is particularly useful for handling cases where multiple effect sizes are extracted from a single study, as it accounts for within-study correlations and prevents overweighting of studies contributing multiple outcomes (Fisher & Tipton, 2015; Tipton, 2015; Tipton & Pustejovsky, 2015). Instead of aggregating effect sizes or selecting only one per study, RVE allows for all reported effects to be included while adjusting for their interdependence through correlated effects modelling (Fisher & Tipton, 2015; Tipton, 2015; Tipton & Pustejovsky, 2015). The Spearman's Rho correlation coefficient (ρ) represents the assumed dependence between effect sizes within a study. Based on prior recommendations (Tipton, 2015), we initially set this value at 0.8 but conducted sensitivity analyses to examine the robustness of our results under different correlation assumptions. Specifically, we varied ρ from 0.1 to 1.0 (Tanner-Smith & Tipton, 2014), assessing whether changes in the assumed within-study dependence led to substantial shifts in the estimated effect sizes. This approach ensures that the summary effects are not unduly influenced by the choice of correlation coefficient and helps verify the stability of our findings.

Between-study heterogeneity was assessed using T^2 to estimate the degree of variability in effect sizes across studies that cannot be attributed to random sampling error. T^2 indicates the variance of the true effect sizes, a higher T^2 value indicates greater heterogeneity, suggesting that true effect sizes vary meaningfully across studies rather than simply due to chance (Deeks et al., 2008; Engels et al., 2000). The square root of this number, Tau (T), estimates the standard deviation of the true effects (Deeks et al., 2008).

To further assess variability, we calculated I^2, which quantifies the proportion of total observed variation that stems from actual differences between studies rather than random error (von Hippel, 2015). I^2 is independent of the number of studies in the meta-analysis a stable (Higgins & Thompson, 2002). Interpretation guidelines suggest that values below 25% indicate low heterogeneity, while values of 50% and 75% suggest moderate and high heterogeneity, respectively (Higgins et al., 2003). Lower I^2 values are preferable, as they indicate that the observed variation is primarily due to chance rather than systematic study differences.

Forest plots were generated for the main effect of ADHD on anxiety symptomology. These show mean effect sizes for each study. The effects in these are underweighted and differ to the main analyses effects as dependencies are not considered as they only include the mean effect size per study and not individual effect sizes.

Moderator Analysis. Separate regressions were used to investigate the effect of moderators. This was done as there was not a complete data set for all moderators. Therefore, if all moderators were entered into one model, the model would exclude an effect size for which any one moderator was missing, therefore many effect sizes would be excluded and the power of the analysis substantially reduced. Individual regressions for each moderator allowed for the case-by-case exclusions of effect sizes based on missing data for the moderator in question only. For categorical variables, an omnibus Wald (HTZ) test was used to test the difference between categories of the moderator (Wald, 1943).

Publication Bias. Several steps were taken to ensure against potential publication bias in line with those utilised by prior meta-analyses (Gunnerud et al., 2020; Hawes et al., 2022). The outcome of these is presented in the results section.

First, published and unpublished studies were compared with moderator analysis. Secondly, funnel plot asymmetry was assessed, following the recommendations of Rodgers and Pustejovsky (2021). Funnel plots visually represent effect sizes (x-axis) in relation to their standard errors (y-axis). Studies with larger standard errors appear toward the bottom of the plot, and the average effect size is shown by a central vertical line. An asymmetrical funnel plot, with a greater number of studies appearing on the right side of the vertical line, may indicate bias (Sterne et al., 2011).

This study utilised two approaches: effect-size-level funnel plots to identify selective outcome reporting (disaggregated effect sizes) and study-level funnel plots to detect selective publication (aggregated effect sizes). An asymmetrical plot with disaggregated effect sizes would indicate selective reporting bias, whereas asymmetrical funnel plots with aggregated effect sizes can suggest publication bias (Sterne et al., 2011). Asymmetry was tested for statistically using Robust Egger's Regression (Egger et al., 1997), specifically utilising the Precision Effect Estimate with Standard Error (PEESE) and the Precision-Effect Test (PET) to analyse disaggregated dependent effects (Rodgers & Pustejovsky, 2021).

The third step taken for publication bias was to use the two-step trim-and-fill method aimed at identifying sources of potential bias and adjusting results for it to indicate the impact of these on effect sizes (Duval & Tweedie, 2000; Mavridis & Salanti, 2014).

Results

Study Characteristics

The meta-analysis on the impact of ADHD on anxiety symptoms included 112 anxiety measurements (effect sizes) from 58 studies from 19 countries. Characteristics of each of the studies are presented in Table 1. In total, 18,821 participants (ADHD $N=5{,}595$; Control $N=13{,}226$) were included in the meta-analysis. The average age for ADHD participants across all studies was 29.11 years ($SD=6.49$), and 29.82 years ($SD=7.13$) for controls. There was a slightly higher average percentage of females across the control groups (57%) than ADHD groups (46.19%). Across studies, the mean number of anxiety measures reported (effect sizes) per study was 1.93 (min = 1, max = 6).

Overall Effect of ADHD on Anxiety

The overall effect of ADHD on anxiety was $g=0.77$, $SE=0.066$, 95% CI [0.638, 0.901], $p<.001$. This indicates a medium effect of ADHD on severity of anxiety symptomology (Cohen, 1988, 1992). Between-study variability, T^2, was estimated to be 0.22, indicating a low to moderate level of variability. Approximately 90.35% of the variance ($I^2=90.47$) related to true variance rather than random error, this indicates substantial heterogeneity (Higgins et al., 2003). Varying the assumed correlation between the within-study effect sizes from 0.1 to 1, had no impact on the overall effect, standard error, and estimates of heterogeneity.

A visual summary of the average effect size (Hedges g) and confidence intervals for each study can be seen in the forest plot (Figure 2). As noted below, this is based on the average effect from each study, yet shows some studies have very wide confidence intervals, potentially suggesting high variability.

Moderator Effects

The results of the moderator analyses are reported in Table 2. Of the five categorical variables examined, only Control Group Type (Clinical vs. Non-clinical) emerged as a statistically significant moderator. A much larger effect was seen when the study utilised a non-clinical control group (mean effect (g) = 0.954) compared to a clinical control group (mean effect (g) = 0.437) ($\beta=-0.517$; $p<.001$). Neither of the continuous variables, age or gender, were significant.

Publication Bias

The effect of ADHD on anxiety severity was different for published studies (mean effect (g) = 0.801) compared to unpublished studies (mean effect (g) = 0.371), but this was not significant ($\beta=-0.429$; $p=.06$) (see Table 2). A Wald test similarly showed no significant effect of publication type as a moderator, F (3.47) = 7.2, $p=.06$, $I^2=90.39$, $T^2=0.22$.

Funnel plots were generated to further assess publication bias. Visual inspection of the plot of aggregated effect sizes (Figure 3a) indicates right sided asymmetry, but the Precision Effect Estimate with Standard Error (PEESE) test was not statistically significant ($p=.055$). A funnel plot of disaggregated effect sizes was generated to consider selective outcome reporting. Visual inspection of this plot (Figure 3b) also showed some right sided asymmetry, and the Precision-Effect Test (PET) test was statistically significant ($p=.05$). The trim and fill method found no indications of missing studies and minimal publication bias.

Discussion

This study addressed the question of whether there is a difference in scores on anxiety measure between individuals with and without ADHD. Prior systematic reviews indicate an increased prevalence of anxiety in individuals with ADHD (Choi et al., 2022; Hartman et al., 2023) however, this is the first meta-analysis quantifying the differences in symptoms and severity of anxiety between adults with and without ADHD, rather than focusing on categorical diagnostic status alone.

The results from this meta-analysis indicate that across studies anxiety scores are higher in adults with ADHD, compared to those without ADHD and that this is a moderate sized effect ($g=0.77$). These findings add to the overall understanding of the relationship between ADHD and anxiety and are important to consider in relation to prior research that has indicated that this co-occurrence can change the presentation and prognosis of each condition (D'Agati et al., 2019).

The meta-analysis indicated some heterogeneity between studies ($T^2=0.22$) with a substantial percentage of this related to true difference rather than random error ($I^2=90.47$)

Table 1. Characteristics of Studies Included in the Meta-Analysis.

Study	Country	*k*	Type	ADHD	Anxiety	ADHD group(s)	Non-ADHD group(s)
Abdel-Hamid et al. (2021)	Germany and Australia	1	C		DÄS-Zuf	*N* = 32, psychiatry (Mean age = 33.75, 50% Female)	*N* = 32, non-clinical control, (Mean age = 33.59, 50% Female)
Ahlers et al. (2022)	Switzerland, Austria and Germany	2	SR	ASRS-V1.1	GAD 7	*N* = 94, ADHD + Cannabis user, pre + post intervention (Mean age = 26.9, 31.9% Female)	*N* = 273, Clinical – Cannabis users, pre + post intervention (Mean age = 28.2, 27.1% Female)
Austin (1998)	USA	4	SR	ADHD BCA	STAI–State STAI–Trait	*N* = 48, psychiatry (Mean age = 34, 50% Female)	*N* = 36, Clinical control (Mean age = 32, 52.78% Female) *N* = 58, non-clinical control (Mean age = 34, 51.72% Female)
Beaton et al. (2022)	UK	1	SR	ASRS-V1.1	HADS	*N* = 543, Mixed (Mean age = 33.56, 63.72% Female)	*N* = 313, non-clinical control, (Mean age = 33.83, 66.45% Female)
Bloch et al. (2017)	Israel	3	C	ASRS-V1.1 WURS	STAI–State STAI–Trait	*N* = 36, Pre & post-medication (Mean age = 30.72, 38.89% Female)	*N* = 25, non-clinical control, Pre & Post-medication (Mean age = 26.84, 88% Female)
Bodalski et al. (2019)	USA	1	SR	BAARS-IV	GAD 7	*N* = 59, Mixed (Mean age = 31.34, 64.41% Female)	*N* = 100, non-clinical control (Mean age = 30.4, 57% Female)
Brona et al. (2018)	The Netherlands	1	C	DIVA-2 ADHD-RS	SCL-90	*N* = 330, Psychiatry (Mean age = 34.9, 45.50% Female)	*N* = 330, non-clinical control (Mean age = 34.6, 51.50% Female)
Bueno et al. (2017)	Brazil	2	C	ASRS-V1.1	STAI–Trait	*N* = 14, Unmedicated (Mean age = 32.4, 50% Female) *N* = 34, Medicated (Mean age = 30.2, 50% Female)	*N* = 20, non-clinical control (Mean age = 28.4, 40% Female)
Canu et al. (2017)	USA	3	SR	WURS	DASS-21 TAI-5 R-MARS	*N* = 40, Mixed (Mean age = 20.1, 62% Female)	*N* = 163, non-clinical control (Mean age = 19.5, 67.90% Female)
Chao et al. (2008)	Taiwan	1	SR	ASRS-V1.1	BAI	*N* = 328, Mixed (Mean age = 22.02, 0% Female)	*N* = 601, non-clinical control (Mean age = 22.41, 0% Female)
Drake et al. (2019)	USA	1	C	BASC-2-C CAARS	BASC-2-SRP-C	*N* = 51, Mixed (Mean age = 19.75, 47.06% Female)	*N* = 34, Clinical control (Mean age = 19.35, 38.24% Female)
Durand and Arbone (2022)	Canada	1	SR	ASRS-V1.1	STAI-6	*N* = 201, Mixed (Mean age = 30.04, 85% Female)	*N* = 206, non-clinical control (Mean age = 26.79, 67% Female)
Ekinci et al. (2013)	Türkiye	2	C	WURS	HARS	*N* = 40, psychiatry (Mean age = 26.5, 48% Female)	*N* = 40, Clinical – Bipolar (Mean age = 25, 45% Female) *N* = 40, non-clinical control (Mean age = 28.4, 42.5% Female)
Evren et al. (2017)	Türkiye	2	SR	ASRS-V1.1	LSAS BAI	*N* = 96, Mixed (Mean age = 21.66, 61.46% Female)	*N* = 359, non-clinical control, (Mean age = 21.8, 59.05% Female)
Graves (2020)	USA	3	C	BAARS-IV	ASEBA	*N* = 156, Mixed (Mean age = 27, 46.80% Female)	*N* = 22, Clinical control (Mean age = 29.8, 63.60% Female) *N* = 49, Clinical – Mood and anxiety disorders (Mean age = 29.6, 61.2-% Female *N* = 107, Clinical – Specific Learning Disability, (Mean age = 29.8, 63.60% Female)
Grogan and Bramham (2016)	Ireland	1	C	CA-ADHD-DI BAARS	HADS	*N* = 160, Psychiatry (Mean age = 28.02, 36.87% Female)	*N* = 92, non-clinical control (Mean age = 28.53, 35.87% Female)
Grogan et al. (2018)	Ireland	4	SR	CAARS	STAI–State STAI–Trait	*N* = 22, ADHD only (Mean age = 30.64, 36.36% Female) *N* = 16, ADHD + Anxiety (Mean age = 31, 56.25% Female)	*N* = 74, non-clinical control (Mean age = 27.64, 71.62% Female) *N* = 14, Clinical – Anxiety, (Mean age = 31.07, 71.43% Female)

(continued)

Table 1. (continued)

Study	Country	*k*	Type	ADHD	Anxiety	ADHD group(s)	Non-ADHD group(s)
Guo et al. (2021)	Germany	4	C	WRI EISRB WURS ASRS-V1.1 CAARS	STAI–State STAI–Trait	*N* = 73, psychiatry (Mean age = 32.4, 35.62% Female)	*N* = 53, Clinical control (Mean age = 39.3, 43.4% Female) *N* = 43, non-clinical control (Mean age = 33.7, 55.81% Female)
Haddad et al. (2009)	UK	1	C	NHIS	STAI–Trait	*N* = 60, ADHD + Tourette's (Mean age = 30.73, 23.40% Female)	*N* = 72, Clinical – Tourette's (Mean age = 34.38, 31.60% Female)
Hirvikoski et al. (2009)	Sweden	1	C	WURS ASRS-V1.1 C-BPRS	BAI	*N* = 28, Unmedicated (Mean age = 33.39, 28.57% Female)	*N* = 28, non-clinical control (Mean age = 33.25, 35.71% Female)
Kim et al. (2006)	Korea	1	C	DISC-IV	BAI	*N* = 28, ADHD + Alcoholic (Mean age = 43.1, 0% Female)	*N* = 57, Clinical – Alcoholic (Mean age = 42.5, 0% Female)
Kim et al. (2017)	Korea	1	SR	AADHD WURS	BAI	*N* = 53, Mixed (Mean age = 21.36, 90.57% Female)	*N* = 320, non-clinical control, (Mean age = 22.95, 81.30% Female)
Kim et al. (2020)	Korea	4	C	ASRS-V1.1	STAI–State STAI–Trait	*N* = 32, ADHD + Depression (Mean age = 21.53, 15.60% Female)	*N* = 32, Clinical – Depression (Mean age = 30.34, 68.8% Female) *N* = 32, non-clinical control (Mean age = 27.87, 59.4% Female)
Kiraz and Sertçelik (2021)	Türkiye	1	C	DIVA-2 ASRS-V1.1	BAI	*N* = 55, ADHD only (Mean age = 22, 54.55% female).	*N* = 52, non-clinical control, (Mean age = 23.55, 63.47% Female)
Kiraz et al. (2021)	Türkiye	1	C	DIVA-2 ASRS-V1.1 WURS	BAI	*N* = 101, psychiatry (Mean age = 23.06, 47.52% Female)	*N* = 100, non-clinical control, (Mean age = 22.7, 50% Female)
Kordahji et al. (2021)	Israel	2	SR	ASRS-V1.1	STAI–State STAI–Trait	*N* = 55, Mixed (Mean age = 27.58, 54.5% Female)	*N* = 35, non-clinical control (Mean age = 32.77, 54.30% Female)
Koyuncu et al. (2015)	Türkiye	1	C		LSAS	*N* = 89, ADHD + Social Anxiety (Mean age = 28.9, 42.69% Female)	*N* = 36, Clinical – Social anxiety (Mean age = 31.7, 33.33% Female)
Lewandowski et al. (2013)	USA	1	SR	ADHD-RS	TTAS	*N* = 35, Mixed (Mean age = 19.71, 40% Female)	*N* = 220, non-clinical control (Mean age = 19.37, 70.91% Female)
Migliarese et al. (2020)	Italy	2	C		STAI–State STAI–Trait	*N* = 9, Psychiatry (Mean age = 28.3, 33.34% Female)	*N* = 50, non-clinical control (Mean age = 29.8, 44% Female) – State *N* = 43, non-clinical control (Mean age = 29, 41.86% Female) – Trait
Morein-Zamir et al. (2021)	UK	1	C	BAARS-IV ASRS-V1.1	DASS	*N* = 88, Psychiatry (Mean age = 32.68, 36.36% Female)	*N* = 90, non-clinical control, (Mean age = 31.99, 35.56% Female)
Mörtberg et al. (2012)	Sweden	2	C	WURS ASRS-V1.1	LSAS	*N* = 178, ADHD only (Mean age = 31, 50% Female)	*N* = 39, Clinical – Social anxiety (Mean age = 35.7, 64% Female) *N* = 88, Clinical control (Mean age = 35.7, 64% Female)
Nazar et al. (2018)	Brazil, UK and The Netherlands	4	C	ASRS-V1.1 K-SADS-PL	STAI–State STAI–Trait	*N* = 35, Unmedicated (Mean age = 24, 80% Female) *N* = 16, ADHD + eating disorder (Mean age = 24, 81.25% Female)	*N* = 39. non-clinical control (Mean age = 23.3, 81.80% Female)
Nelson and Liebel (2018)	USA	1	C		BASC-2-SRP-C	*N* = 150, Mixed (Mean age = 19.22, 50% Female)	*N* = 150, non-clinical control (Mean age = 19.15, 50% Female)
Nelson et al. (2014)	USA	1	SR		TAI-5	*N* = 60, Mixed (Mean age = 19.85, 41.67% Female	*N* = 60, non-clinical control (Mean age = 18.8, 41.67% Female)

(continued)

Table 1. (continued)

Study	Country	*k*	Type	ADHD	Anxiety	ADHD group(s)	Non-ADHD group(s)
O'Rourke et al. (2020)	USA	3	C	ADHD-RS CAARS	BAI LSAS PSWQ	*N* = 46, Mixed, (Mean age = 20.5, 67% Female)	*N* = 46, non-clinical control, (Mean age = 20.0, 65% Female)
Oh et al. (2018)	Korea	1	C	MINI CAARS	STAI–Trait	*N* = 51, Psychiatry (Mean age = 27.82, 35.30% Female)	*N* = 51, non-clinical control, (Mean age = 24.33, 35.30% Female)
Özten et al. (2016)	Türkiye	2	C	T-ADHD-SRS	A-SAQ SASI	*N* = 30, Psychiatry (Mean age = 34.3, 40% Female)	*N* = 26, Clinical control (Mean age = 36.9, 38.50% Female)
Pehlivanidis et al. (2014)	Greece	3	C		STAI–State STAI–Trait SCL-90	*N* = 22, ADHD + comorbidity (Mean age = 33.8, 31.82% Female)	*N* = 92, Clinical – anxiety and depression (Mean age = 37.8, 63.04% Female)
Pinnaa et al. (2019)	Italy, Canada and USA	1	SR	ASRS-V1.1	HARS	*N* = 173, ADHD + Bipolar (Mean age = 46, 34.11% Female)	*N* = 530, Clinical – Bipolar (Mean age = 46, 61.69% Female)
Quintero et al. (2019)	Spain	6	C	DIVA-2 WURS CAARS	STAI–State STAI–Trait	*N* = 31, ADHD only (Mean age = 41.7, 54.84% Female) *N* = 31, ADHD + comorbidity (Mean age = 37.8, 62.29% Female) *N* = 29, ADHD + comorbidity + childhood diagnosis (Mean age = 30.5, 51.72% Female)	*N* = 25, non-clinical control (Mean age = 43.6, 56% Female)
Raj et al. (2009)	USA	4	C	CAARS	BAI ASI	*N* = 18, Unmedicated (Mean age = 41.8, 55.56% Female)	*N* = 21, Clinical – postural tachycardia syndrome (Mean age = 35.9, 86% Female) *N* = 20, non-clinical control (Mean age = 36.6, 90% Female)
Rodriguez (2007)	USA	2	SR		STAI–State STAI–Trait	*N* = 37, Mixed (Mean age = 20.78, 0% Female)	*N* = 90, non-clinical control (Mean age = 19.78, 0% Female) – State *N* = 88, non-clinical control (Mean age = 19.78, 0% Female) – Trait
Rogers et al. (2017)	UK	2	C	BCSS	HADS	*N* = 204, Psychiatry (Mean age = 36, 30% Female)	*N* = 86, Clinical – Chronic fatigue (Mean age = 36, 73% Female) *N* = 243, non-clinical control (Mean age = 31, 73% Female)
Roth et al. (2004)	USA	1	C	WURS	STAI–State	*N* = 28, unmedicated (Mean age = 34.2, gender not specified)	*N* = 34, non-clinical control (Mean age = 34, gender not specified)
Rucklidge et al. (2007)	UK and Canada	2	SR	CAARS ADDC	BAI	*N* = 52, Mixed (Mean age = 39.8, 100% Female) *N* = 37, Mixed (Mean age = 37, 0% Female)	*N* = 51, non-clinical control (Mean age = 41.5, 100% Female) *N* = 40, non-clinical control (Mean age = 44.8, 0% Female)
Şahan et al. (2021)	Türkiye	1	C	SCID-V WURS	BAI	*N* = 55, ADHD + Obese (Mean age = 35.59, 78% Female)	*N* = 55, Clinical – Obese (Mean age = 34.46, 66% Female)
Sertçelik et al. (2022)	Türkiye	1	C	WURS ASRS-V1.1 DIVA-2	BAI	*N* = 158, Unmedicated (Mean age = 22.63, 53.16% Female)	*N* = 96, Clinical control (Mean age = 22.68, 57.29% Female)
Shakeshaft et al. (2023)	UK	4	SR	SDQ	GAD 7	*N* = 224, Mixed, pre-pandemic (Mean age = 21, 62% Female) *N* = 244, Mixed, COVID 1 (Mean age = 27.5, 62% Female) *N* = 241, Mixed, COVID 2 (Mean age = 27.5, 62% Female) *N* = 382, Mixed, COVID 4 (Mean age = 27.5, 62% Female)	*N* = 1,774, non-clinical control, pre-pandemic (Mean age = 21, 70% Female) *N* = 1,804, non-clinical control, COVID 1 (Mean age = 27.5, 70% Female) *N* = 1,656, non-clinical control, COVID 2 (Mean age = 27.5, 70% Female) *N* = 2,646, non-clinical control, COVID 3 (Mean age = 27.5, 70% Female)

(continued)

Table 1. (continued)

Study	Country	*k*	Type	ADHD	Anxiety	ADHD group(s)	Non-ADHD group(s)
Strand (2016)	USA	2	SR		GAD 7	*N* = 134, ADHD + Comorbidity (Mean age = 21, 56% Female)	*N* = 2,950, non-clinical control (Mean age = 20, 54.70% Female) *N* = 708, Clinical – anxiety and depression (Mean age = 21, 63% Female)
Torrente et al. (2014)	Argentina	2	C	BAARS	STAI–State STAI–Trait	*N* = 35, Psychiatry (Mean age = 31.2, 43.86% Female)	*N* = 20, Clinical control (Mean age = 33.3, 50% Female) *N* = 20, non-clinical control (Mean age = 33.8, 90% Female)
Torrente et al. (2017)	Argentina	2	C	ADHD-RS	STAI–Trait	*N* = 60, Psychiatry (Mean age = 34.68, 31.67% Female)	*N* = 50, Clinical – Bipolar (Mean age = 37.02, 48% Female) *N* = 30, non-clinical control (Mean age = 35.69, 53.33% Female)
Umutlu et al. (2022)	Türkiye	1	C	K-SADS-PL WURS T-ADHD-SRS	LSAS	*N* = 32, ADHD + Social Anxiety (Mean age = 27.1, 37.5% Female)	*N* = Clinical – Social anxiety (Mean age = 26.4, 42.5% Female)
Weyandt et al. (2023)	USA	2	C	ADHD-RS	BAI	*N* = 105, Unmedicated (Mean age = 18.29, 54.3% female). *N* = 99, Medicated (Mean age = 18.22, 54.5% female).	*N* = 217, non-clinical control, (Mean age = 18.2, 52.1% female).
Yoldas et al. (2019)	Türkiye	1	C	T-ADHD-SRS WURS	LSAS	*N* = 34, ADHD + Social Anxiety (Mean age = 25.3, 50% Female)	*N* = 42, Clinical – Social anxiety (Mean age = 25.5, 43.30% Female)
Young and Gudjonsson (2006)	UK	2	C	YAQ ADHD-AFI	HADS	*N* = 83, Psychiatry (Mean age = 29.8, 27.71% Female)	*N* = 64, Clinical control (Mean age = 32.9, 34.37% Female) *N* = 33, non-clinical control (Mean age = 25, 33.33% Female)
Young et al. (2003)	UK	2	C	Conners 10 ADHD-AFI	HADS	*N* = 30, Psychiatry (Age and gender not specified)	*N* = 30, Clinical control (Age and gender not specified) *N* = 30, non-clinical control (Age and gender not specified)
Yücens et al. (2021)	Türkiye	1	C	WURS T-ADHD-SRS	BAI	*N* = 31, ADHD + OCD (Mean age = 29.6, gender not specified)	*N* = 166, Clinical – OCD (Mean age = 31.2, gender not specified)
Zhang et al. (2021)	China	2	C	ADHD-RS	SAS	*N* = 119, ADHD only (Mean age = 26.3, 24.37% Female) *N* = 119, ADHD with comorbidity (Mean age = 26.7, 46.15% Female)	*N* = 54, non-clinical control (Mean age = 25.7, 38.89% Female).

Note. *N* = number of participants within group. *k* = number of effect sizes in the study; Type = method of categorising ADHD with C = clinical and SR = self-report; ADHD = measures used to assess ADHD; Anxiety = measures used to assess anxiety; see Appendix C for full scale names and information; see Appendix D for full scale names and information.

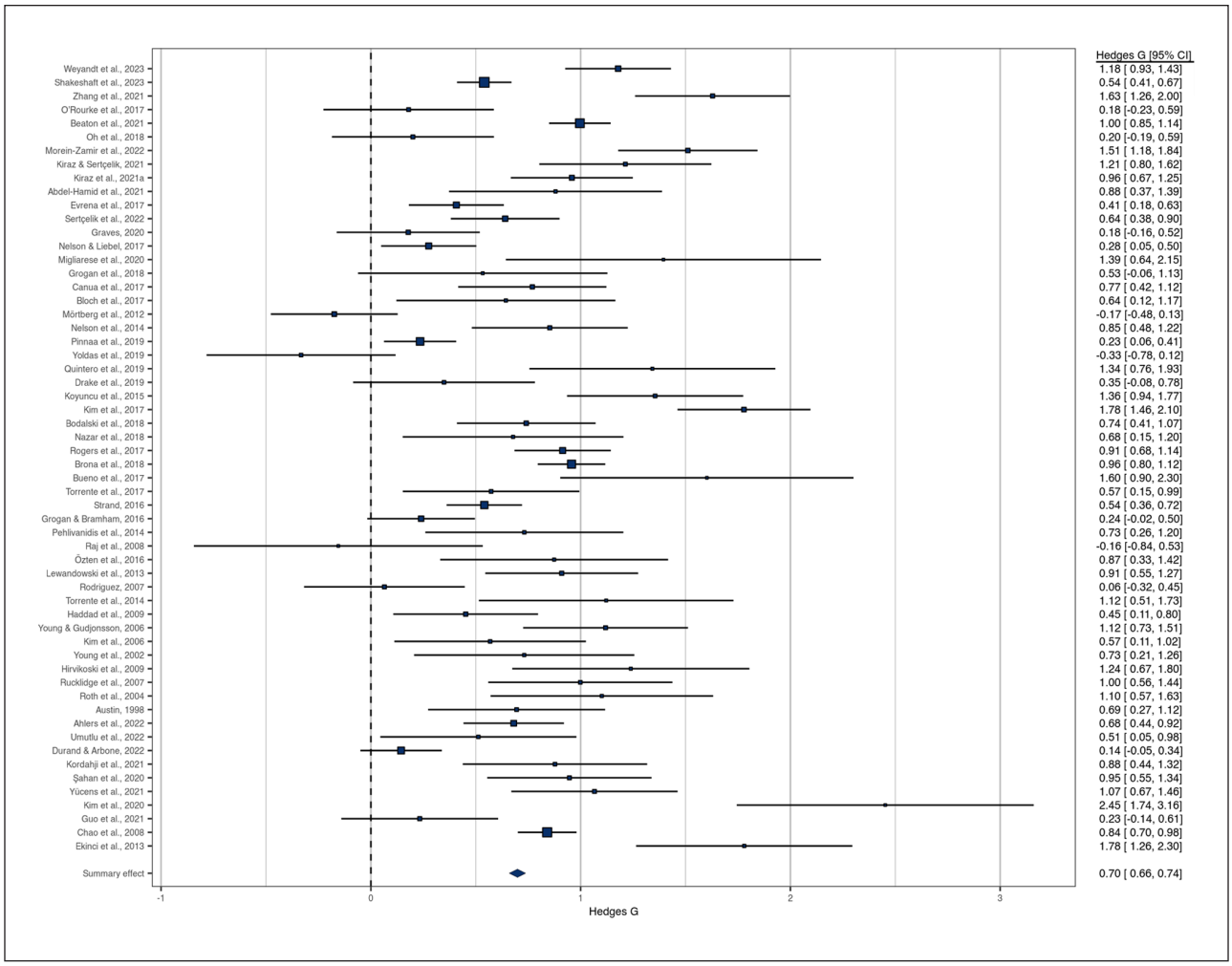


Figure 2. Forest plot of effect sizes of differences in anxiety scores between those with and without ADHD scores.

(Higgins et al., 2003). Heterogeneity is important within meta-analyses as it can indicate that studies are highly different and not measuring the same effect. Combining these studies could be unsuitable (Engels et al., 2000) and present challenges in drawing a robust overall conclusion. Although I^2 is independent to the number of studies included in a meta-analysis (Higgins & Thompson, 2002), research has indicated that the I^2 statistic increases with larger samples (Rücker et al., 2008). This indicates a limitation in using the I^2 statistic, especially using it to indicate clinical heterogeneity in studies (Borenstein et al., 2017; Rücker et al., 2008). The T^2 value may be more representative of variation between studies as this is insensitive to sample size or number of studies (Rücker et al., 2008). There was a low T^2 in this study, indicating low to moderate heterogeneity. Furthermore, as studies included were expected to be diverse yet clinically relevant to the research question, the heterogeneity reinforced the need for the moderator analysis.

Moderators in the Relationship Between ADHD and Anxiety Symptomology

Interestingly, of the moderators considered, only the type of control group emerged as having a statistically significant effect. When a non-clinical population was used as the comparison group, a large mean effect ($g=0.954$) was seen (Cohen, 1988, 1992). This is in line with findings that in the general population there is a higher prevalence of anxiety disorders in individuals with ADHD (Choi et al., 2022; Kessler et al., 2006) and provides further evidence for the suggestion that individuals with ADHD experience greater levels of anxiety (Agarwal et al., 2012; Nelson & Liebel, 2018).

When adults with ADHD were compared to a clinical population without ADHD a small to medium mean effect ($g=0.437$) was seen (Cohen, 1988, 1992). This indicates that although the prevalence rate of anxiety is higher in individuals from psychiatric populations with ADHD than without (Choi et al., 2022) there is less difference in the

Table 2. Results of Moderator Analyses for the Effect of ADHD on Anxiety.

			Heterogeneity			Treatment effects					
Moderator	*m*	*k*	I^2	T^2	*F*	β (Hedges's *g*)	*SE*	95% CI	*t*	*df*	*p*
Categorical variable moderators											
ADHD diagnosis type	58	112	90.63	0.23	0.26					30.90	.617
Clinical	41	78				0.794	0.08	[0.62, 0.96]	9.44	40.50	<.001
Self-report	17	34				0.726	0.09	[0.52, 0.92]	7.77	15.90	<.001
Type of control group	58	112	89.02	0.19	16.2					43.50	<.001
Non-clinical	51	86				0.954	0.08	[0.78, 1.12]	11.30	38.10	<.001
Clinical	21	29				0.437	0.09	[0.23, 0.64]	4.45	22.80	<.001
Anxiety measure type	58	112	90.43	0.22	2.37					13.00	.15
Overall anxiety	63	96				0.816	0.07	[0.68, 0.95]	11.86	47.91	<.001
Specific anxiety	14	16				0.527	0.18	[0.13, 0.92]	2.99	9.32	.01
Anxiety focus: reference – state	55	108	90.33	0.23	0.09					37.1	.768
Anxiety focus: reference – trait	55	108	90.33	0.23	0.23					34.8	.634
State	29	42				0.767	0.12	[0.52, 1.01]	6.46	23.1	<.001
Prolonged state	19	29				0.812	0.09	[0.61, 1.01]	8.50	16.8	<.001
Trait	26	37				0.767	0.09	[0.49, 0.90]	7.20	19.2	<.001
Publication status	58	112	90.39	0.22	7.2					3.47	.06
Peer reviewed	54	101				0.801	0.07	[0.66, 0.94]	11.69	52.30	<.001
Dissertation	4	11				0.371	0.14	[−0.09, 0.83]	2.57	2.99	.08
Continuous variable moderators											
Age of ADHD (years)	57	110	90.7	0.23		−0.004	0.01	[−0.02, 0.02]	−0.42	19.30	.68
% of females ADHD	55	108	90.97	0.24		−0.029	0.39	[−0.88, 0.82]	−0.07	13.4	.94

Note. *k* = number of effect sizes; *m* = number of studies for each moderator; I^2 = true heterogeneity; T^2 = variation in effect sizes between studies; 95% CI = 95% confidence interval; *df* = degrees of freedom. For the analyses of categorical moderators, β values indicate the effect size (Hedges's *g*) for each level of the moderator compared against zero; *F* = *F*-value associated with an omnibus Wald (HTZ) test, used to test the null hypothesis that the average effect size is the same across all levels of the moderator; For the analyses of the continuous moderators (bottom of table), β values represent the slope coefficient estimates and are a measure of the effect (Hedges's *g*) of each moderator against the intercept; *p* = *p*-value of the *t*-tests that compare each moderator against the intercept. *p* = *p*-value of HTZ Wald tests for overall moderator effects and *p*-value of *t*-tests that compare each level of each moderator against zero. Where the degrees of freedom are less than 4, there is a high chance of Type 1 error and results should be viewed as unreliable.

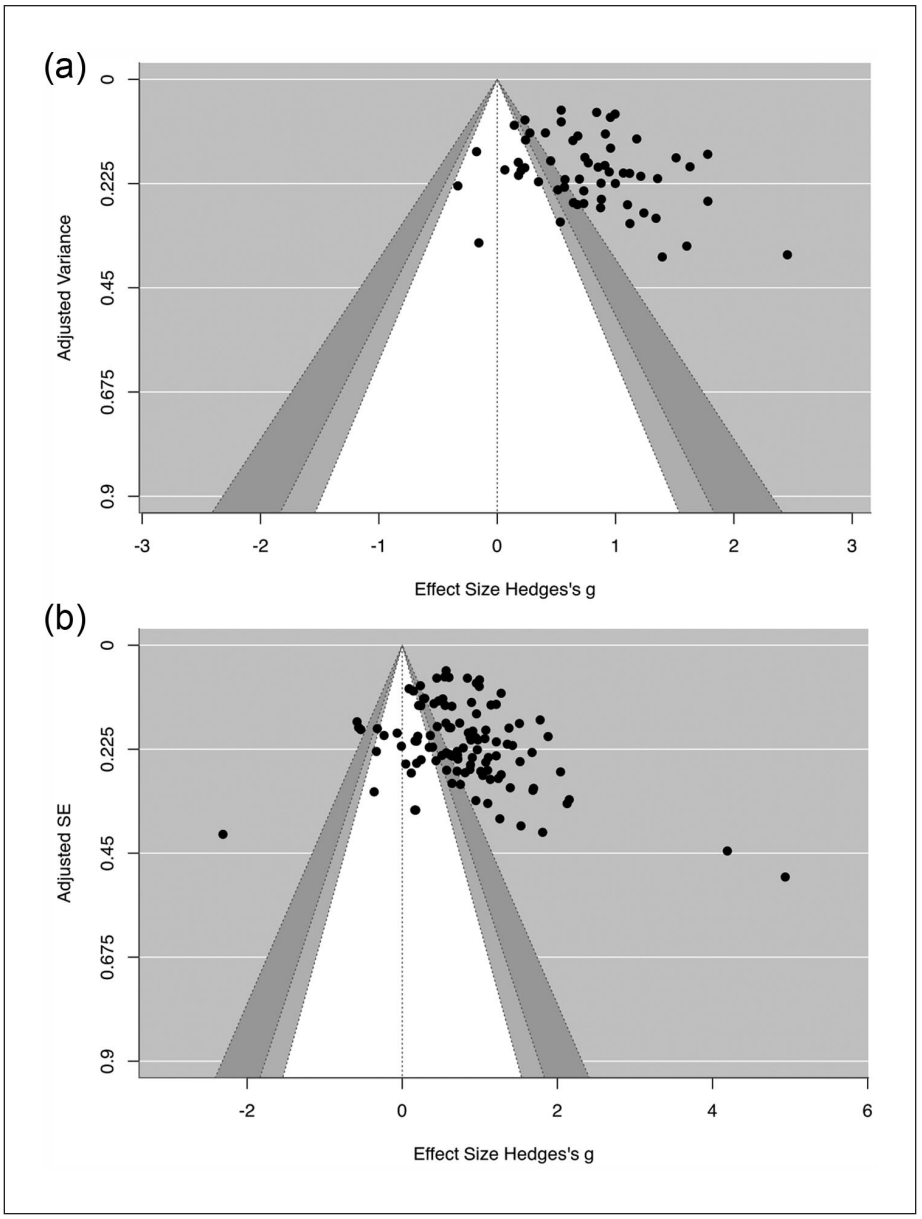


Figure 3. (a) Study level funnel plot of aggregated effect sizes for ADHD on anxiety and (b) effect size level funnel plot of disaggregated effect sizes for ADHD on anxiety.

severity of anxiety symptomology. This result may relate to findings that anxiety and mood disorders are the dominant conditions that individuals seek mental health treatment for (Gorlin et al., 2016; Wang et al., 2005), for example in some of these studies the comparisons made are very conservative comparing adults with ADHD to adults with a clinical diagnosis of anxiety or depression.

There are some potential limitations of the binary method used in this study to categorise control groups. This approach was selected due to several studies having unspecified mixed clinical control groups. Future research could consider how the specific nature of a clinical control group moderates the effect of ADHD on anxiety symptoms. Particularly as negative effects were seen when individuals with ADHD were compared to individuals from specific sub-groups, for example those with Postural orthostatic tachycardia syndrome (POTS; Raj et al., 2009).

Furthermore, studies that included a clinical control group with anxiety were included in this study. This was due to the overarching interest in considering whether the presence of ADHD impacted on anxiety symptom severity when compared to any non-ADHD population. A further

level to the control group moderator was considered to differentiate clinical control groups which included those with anxiety, however, this was not possible as there were too few studies to split the effect sizes and the analysis would have been under powered. It would be useful for future meta-analyses to compare scores on anxiety measures for individuals with anxiety with and without co-occurring ADHD.

Although this study found that the type of anxiety measure used (general vs. specific) did not moderate the effect of ADHD on anxiety scores, the "specific" category encompassed various forms of anxiety, such as test anxiety and separation anxiety. Given the diverse nature of anxiety disorders, future research should investigate whether differences in anxiety scores between individuals with and without ADHD vary depending on what specific type of anxiety is assessed. For example, despite previous research indicating a higher prevalence of social anxiety in individuals with ADHD (Gorlin et al., 2016; Kessler et al., 2006), two studies in this meta-analysis reported a negative effect of ADHD on LSAS (Liebowitz, 1987) scores when comparing individuals with social anxiety to those with ADHD (Mörtberg et al., 2012) or both ADHD and social anxiety (Yoldas et al., 2019).

Despite prior research highlighting the impact of heterogeneity in ADHD diagnosis and definition on the co-occurrence between ADHD and anxiety diagnoses (Choi et al., 2022; Koyuncu et al., 2022), no moderation effect was found for the method used to categorise the ADHD group. Little difference was seen in the effect sizes when a study used a clinical procedure including an interview ($g = 0.794$), or an ADHD questionnaire or self-report alone ($g = 0.726$). The criteria for inclusion in each category was agreed by the primary research team, yet this relied on how the authors of studies described the method used to determine ADHD. A lack of detail regarding the ADHD classification criteria used in some papers may impact on the validity of the categories used in this analysis. It will be important for authors of future research to ensure they include clear descriptions of how ADHD status is determined to facilitate the inclusion of this as a potential moderating factor in meta-analyses on ADHD. Future research may also wish to include correlational analysis comparing the severity of ADHD and anxiety measures.

Neither age or gender of the ADHD group were found to influence the effect of ADHD on anxiety symptomology. However, some studies did not specify age or gender or only did so for the whole sample. As this study focused on adults, the mean age of participants within studies ranged from 18.2 to 46 years. It would be beneficial for future research to include under 18s and adults later in life to better understand the relationship between co-occurring ADHD and anxiety across the full life span. There were more males in the ADHD groups which is in line with findings that females are often under-diagnosed with ADHD (Adamis et al., 2022; Robison et al., 2008). Despite research indicating a higher prevalence of ADHD in Transgender and Gender Diverse (TGD) individuals (Goetz & Adams, 2024), only one study found within this review reported the number of TGD participants (Bodalski et al., 2019). Future research is needed to further understand the symptomology and severity of co-occurring ADHD and anxiety across gender identities, particularly given the risk of higher minority stress in TGD individuals (Ignatova et al., 2025).

This review included studies from around the world (Table 1), with the highest proportion originating from Europe ($N = 21$, 36.21%). The individual countries with the largest contribution were the United States of America ($N = 16$, 27.59%) and Türkiye ($N = 11$, 18.96%). A small proportion of studies were from Asia ($N = 6$, 10.34%), and South America ($N = 4$, 6.90%) but no studies were found from Africa. Although there has been an increase in global awareness and interest in ADHD, geographical differences have been found in the focus and publication of ADHD research (Hodgkins et al., 2012). As cultural differences have been seen in the expression and recognition of both ADHD and anxiety (Hofmann & Hinton, 2014; Wilcox et al., 2007), it would be beneficial for future research to explore whether this influences the relationship between ADHD and co-occurring anxiety.

Considerations

The findings of this meta-analysis should be interpreted in the context of some limitations that may have influenced the effects. First, only one author extracted data from the studies for analysis. However, steps were taken to ensure data accuracy, including reviewing extracted data prior to creation of effect sizes, then inspecting any outliers. Furthermore, criteria for allocation to moderator categories was agreed within the research team. Although there is evidence indicating that independent data extraction can result in more errors (Buscemi et al., 2006), errors and disagreements have similarly been seen when multiple reviewers are utilised, yet little impact has been found on overall conclusions (Jones et al., 2005).

Although no significant moderation effect was found for publication type suggesting lower risk of publication bias, as the degrees of freedom were less than four, this increases the likelihood of a type 1 error (Tanner-Smith et al., 2016). The lack of significance may be due to the small number of studies within the dissertation group. Overall, there does appear to be some publication bias within these findings, so results should be interpreted with this in mind. To aid with this, future researchers should endeavour to adhere to open research practices and to publish/make available all findings on anxiety and ADHD, even if not significant, that is, null findings.

Implications

Anxiety is a common sometimes debilitating condition (Davies et al., 2022) and has been seen to worsen ADHD symptomology (Reimherr et al., 2017) and have a significant impact on the quality of life of individuals with ADHD (Agarwal et al., 2012). Despite this, there is a lack of awareness regarding the association and interaction between these co-occurring conditions (D'Agati et al., 2019). The findings of this study provide further insight into anxiety symptomology for individuals with ADHD and how this can differ in severity to individuals without ADHD. This information is important for healthcare professionals who are involved in the assessment or treatment of individuals with ADHD and/or anxiety, as this deeper understanding will allow for appropriate adjustments to practice and guidelines. This may include further assessment questions to differentiate symptomology, or screening for anxiety in individuals with ADHD and providing resources and signposting for symptom management so that co-occurring anxiety is not overlooked. Given that ADHD is often underdiagnosed in individuals with other co-occurring mental health difficulties (Gorlin et al., 2016), these findings also reinforce the suggestion to test adult patients presenting with more severe anxiety symptomology for ADHD (Pehlivanidis et al., 2014). Importantly, it will be important for clinicians to distinguish symptoms carefully to prevent misdiagnosis (Koyuncu et al., 2022). Such adjustments could lead to substantial improvements in the quality of life of individuals with ADHD and reduce the likelihood of further psychiatric difficulties developing (Brown, 2000; Katzman et al., 2017).

The findings of this study also have implications for individuals with ADHD and their support network. For example, psychoeducation on anxiety symptomology in ADHD could lead to better understanding and reduced stigma from self or others. These preventative measures may also support children with ADHD who may be at risk of elevated anxiety. However, further research is needed to understand if there is a difference across the lifespan. Furthermore, additional meta-analysis research into the impact of anxiety on ADHD symptomology would be beneficial due to potential implications for the effective treatment of ADHD (Gorlin et al., 2016) given recent debate as to whether anxiety exacerbates ADHD symptomology or is a protective factor (Jarrett & Ollendick, 2008; Reimherr et al., 2017).

Conclusion

This study aimed to comprehensively and systematically review the literature to better understand the relationship between ADHD and anxiety. Analysis found that individuals with ADHD score higher on measures of anxiety than individuals without ADHD. This adds to prior research indicating an increased prevalence of anxiety disorders in individuals with ADHD, indicating that anxiety symptoms are elevated in those with ADHD irrespective of anxiety diagnostic status. This effect is moderated by the type of comparison group used, with the difference being larger when individuals with ADHD are compared to a non-clinical population than to a clinical population. No moderation effect was found from the age or gender ratio of the ADHD group, the method of classifying ADHD or the type or focus of the anxiety measure used. These findings have important implications for the diagnosis and treatment of individuals with ADHD and co-occurring anxiety to ensure effective support for this population to improve symptom management and overall quality of life. Future research is needed to better understand factors influencing the relationship between ADHD and anxiety to help towards comprehending the aetiology and chronicity of this co-occurrence.

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Ethical Considerations

This review article does not contain any studies with human or animal participants. Therefore, no ethical review was required.

Consent to Participate

As this review article did not contain participants, no informed consent was required.

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Data Availability Statement

All data and annotated analyses with the accompanying R code can be accessed on our OSF page (https://osf.io/rh6du/).

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Author Biographies

Jessica I. Hoskins has recently completed the Clinical Psychology Doctorate at the University of Surrey. Her doctoral research focused on areas related to ADHD in adulthood, considering the overlap with mental health and what leads people to seek a diagnosis. This interest is also increased by her personal lived experience of receiving a diagnosis of ADHD within adulthood. She has previously completed qualitative and quantitative research within Eating Disorders and service user experiences.

Debbie Gooch is a Lecturer in Developmental Psychology at the University of Surrey and co-director of the Cognition, Genes and Developmental Variability (CogDev) Lab. Her research focuses on understanding variability in the development of language, literacy, attention, and executive function using a range of methodologies. Specifically, she is interested in understanding why some individuals experience persistent difficulties in these areas, why neurodevelopmental conditions such as developmental language disorder, dyslexia and ADHD frequently co-occur and how cognitive and environmental factors shape the experiences, academic achievement, and wellbeing of individuals with neurodevelopmental conditions.

Katie A. Gilligan-Lee is an expert in developmental psychology and has experience working with both individuals with typical development and neurodivergent groups. Dr Gilligan-Lee has experience in recruiting and testing hard to reach groups, and in the interview methodology and analyses that will be required for this project.

Appendix A: Search Strings

Psych Info/Medline/ERIC via EBSCOhost

TI (“Attention Deficit Disorder” OR “Attention Deficit Hyperactivity Disorder” OR “ADHD” OR “Hyperkinetic Syndrome” OR “Minimal Brain Dysfunction) AND (“Anxiety” OR “Phobia” OR “Sociophobia” OR “Panic” OR “Agoraphobi*” OR “Claustrophobi*” OR “GAD” OR “Hypochondria”)
OR
AB (“Attention Deficit Disorder” OR “Attention Deficit Hyperactivity Disorder” OR “ADHD” OR “Hyperkinetic Syndrome” OR “Minimal Brain Dysfunction”) AND (“Anxiety” OR “Phobia” OR “Sociophobia” OR “Panic” OR “Agoraphobi*” OR “Claustrophobi*” OR “GAD” OR “Hypochondria”)

Scopus

TITLE-ABS ((“Attention Deficit Disorder”) OR (“Attention Deficit Hyperactivity Disorder”) OR (adhd) OR (“Hyperkinetic Syndrome”) OR (“Minimal Brain Dysfunction”) AND (anxiety) OR (phobia) OR (sociophobia) OR (panic) OR (agoraphobi*) OR (claustrophobi*) OR (gad) OR (hypochondria) AND NOT (child*) AND NOT (“young person*”) AND NOT (adolesc*) AND NOT (teen*) AND NOT (youth*) AND NOT (infant*)) AND PUBYEAR > 1993 AND PUBYEAR < 2025 AND (LIMIT-TO (DOCTYPE , “ar”)) AND (LIMIT-TO (LANGUAGE , “English”))

Proquest

abstract((“Attention Deficit Disorder”) OR (“Attention Deficit Hyperactivity Disorder”) OR (adhd) OR (“Hyperkinetic Syndrome”) OR (“Minimal Brain Dysfunction”)) AND abstract((anxiety) OR (phobia) OR (sociophobia) OR (panic) OR (agoraphobi*) OR (claustrophobi*) OR (gad) OR (hypochondria)) OR title((“Attention Deficit Disorder”) OR (“Attention Deficit Hyperactivity Disorder”) OR (adhd) OR (“Hyperkinetic Syndrome”) OR (“Minimal Brain Dysfunction”)) AND title((anxiety) OR (phobia) OR (sociophobia) OR (panic) OR (agoraphobi*) OR (claustrophobi*) OR (gad) OR (hypochondria)) NOT title((child*)) NOT title((“young person*”)) NOT title((adolesc*)) NOT title((teen*)) AND title((youth*)) AND title((infant*))
NOT **TI** (child* OR “young person*” OR adolesc* OR teen* OR youth* OR infant*)
NOT MA (child* OR “young person*” OR adolesc* OR teen* OR youth* OR infant*)

Appendix B: Analytical Approach

Outputs from Rstudio demonstrating each stage of analysis (Figures B1–B13).

ADHD Main Effect

```
RVE: Correlated Effects Model with Small-Sample Corrections

Model: hedges ~ 1

Number of studies = 58
Number of outcomes = 112 (min = 1 , mean = 1.93 , median = 2 , max = 6 )
Rho = 0.8
I.sq = 90.47352
Tau.sq = 0.2247707

               Estimate StdErr t-value  dfs P(|t|>) 95% CI.L 95% CI.U Sig
1 X.Intercept.     0.77 0.0657    11.7 56.3       0    0.638    0.901 ***
---
Signif. codes: < .01 *** < .05 ** < .10 *
```

Figure B1. Main effect output with rho set at 0.8.

```
Number of studies = 58
Number of outcomes = 112 (min = 1 , mean = 1.93 , median = 2 , max = 6 )
Rho = 0.1
I.sq = 90.44355
Tau.sq = 0.2246962

               Estimate StdErr t-value  dfs P(|t|>) 95% CI.L 95% CI.U Sig
1 X.Intercept.     0.77 0.0657    11.7 56.3       0    0.638    0.901 ***
---
Signif. codes: < .01 *** < .05 ** < .10 *
---
```

Figure B2. Main effect output with rho set at 0.1.

```
Number of studies = 58
Number of outcomes = 112 (min = 1 , mean = 1.93 , median = 2 , max = 6 )
Rho = 1
I.sq = 90.48209
Tau.sq = 0.224792

               Estimate StdErr t-value  dfs P(|t|>) 95% CI.L 95% CI.U Sig
1 X.Intercept.     0.77 0.0657    11.7 56.3       0    0.638    0.901 ***
---
Signif. codes: < .01 *** < .05 ** < .10 *
```

Figure B3. Main effect output with rho set at 1.0.

Moderator Analysis

```
Model: HedgesG ~ Age

Number of studies = 57
Number of outcomes = 110 (min = 1 , mean = 1.93 , median = 2 , max = 6 )
Rho = 0.8
I.sq = 90.70914
Tau.sq = 0.2340197

              Estimate  StdErr t-value  dfs P(|t|>) 95% CI.L 95% CI.U Sig
1 X.Intercept.  0.88857 0.30538   2.910 21.6 0.00822   0.2546   1.5225 ***
2          Age -0.00412 0.00993  -0.415 19.3 0.68305  -0.0249   0.0166
---
Signif. codes: < .01 *** < .05 ** < .10 *
```

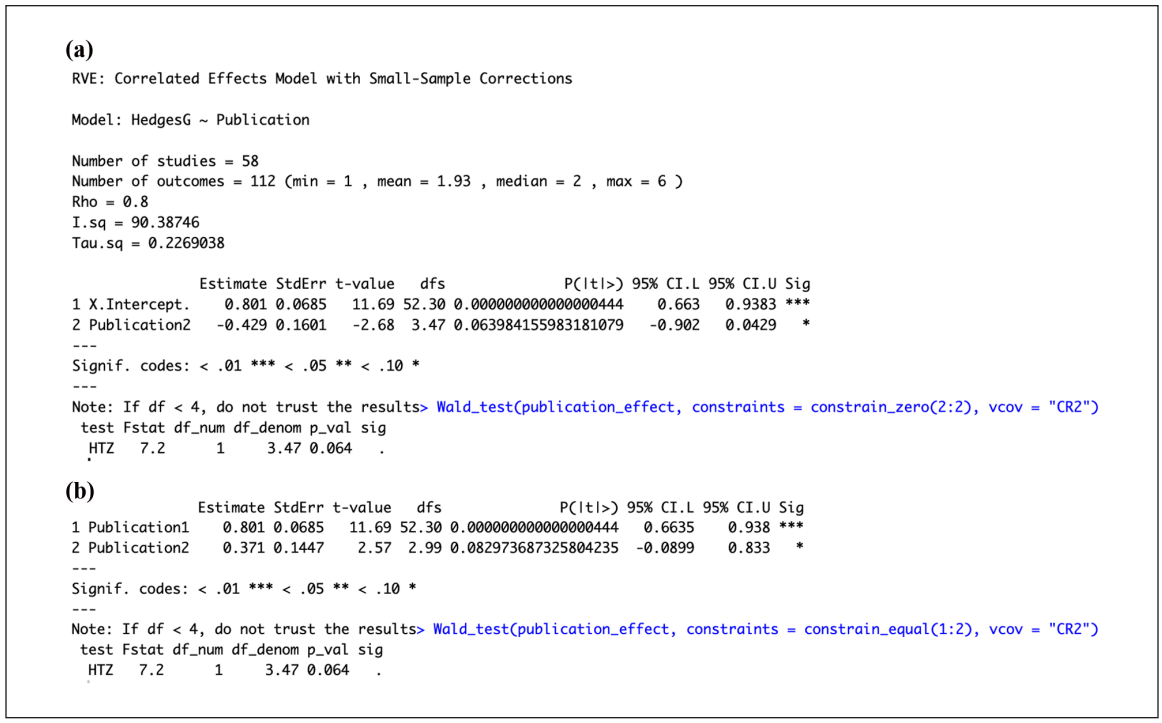
Figure B4. Moderator analysis for age.

```
Model: HedgesG ~ gender

Number of studies = 55
Number of outcomes = 108 (min = 1 , mean = 1.96 , median = 2 , max = 6 )
Rho = 0.8
I.sq = 90.96984
Tau.sq = 0.2392645

              Estimate StdErr t-value  dfs P(|t|>) 95% CI.L 95% CI.U Sig
1 X.Intercept.   0.7737  0.191  4.0591 14.2 0.00114    0.365    1.182 ***
2       gender  -0.0294  0.396 -0.0741 13.4 0.94201   -0.882    0.824
---
Signif. codes: < .01 *** < .05 ** < .10 *
```

Figure B5. Moderator analysis for gender.



(a)

```
RVE: Correlated Effects Model with Small-Sample Corrections

Model: HedgesG ~ Publication

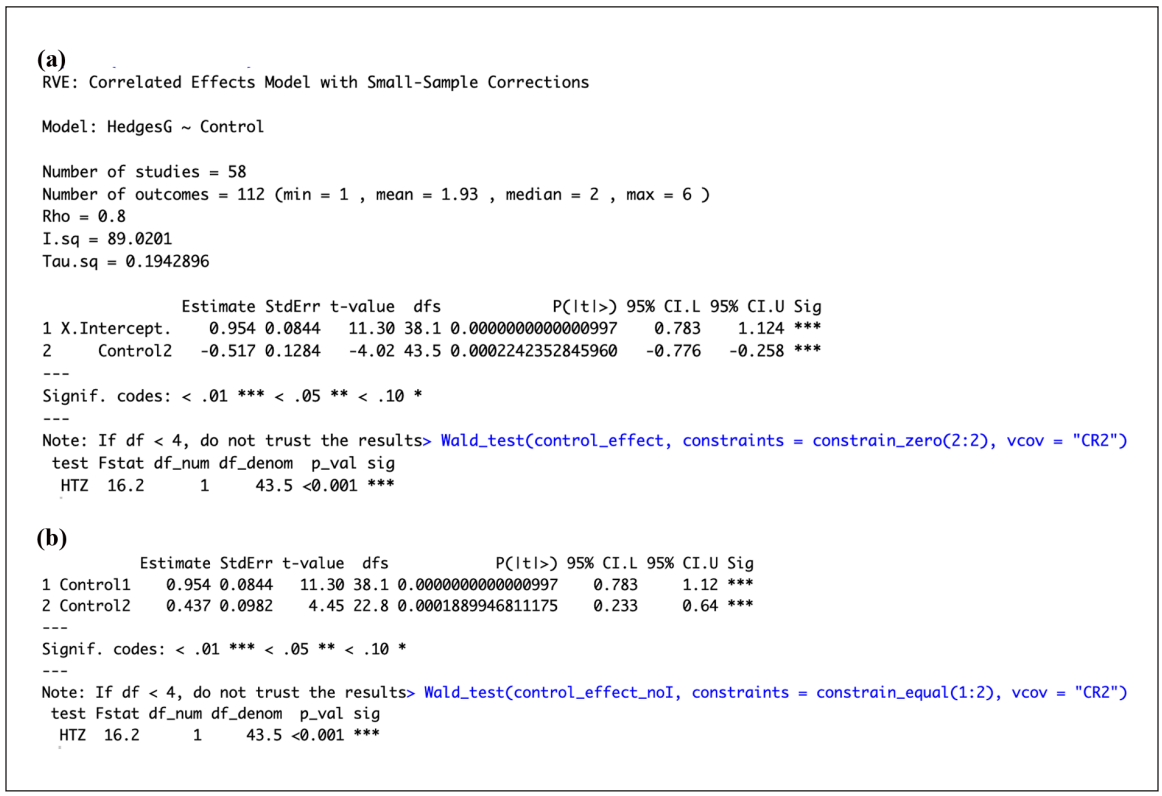
Number of studies = 58
Number of outcomes = 112 (min = 1 , mean = 1.93 , median = 2 , max = 6 )
Rho = 0.8
I.sq = 90.38746
Tau.sq = 0.2269038

               Estimate StdErr t-value   dfs              P(|t|>) 95% CI.L 95% CI.U Sig
1 X.Intercept.    0.801 0.0685   11.69 52.30 0.000000000000000444    0.663   0.9383 ***
2 Publication2   -0.429 0.1601   -2.68  3.47 0.063984155983181079   -0.902   0.0429   *
---
Signif. codes: < .01 *** < .05 ** < .10 *
---
Note: If df < 4, do not trust the results> Wald_test(publication_effect, constraints = constrain_zero(2:2), vcov = "CR2")
 test Fstat df_num df_denom p_val sig
  HTZ   7.2      1     3.47 0.064   .
```

(b)

```
               Estimate StdErr t-value   dfs              P(|t|>) 95% CI.L 95% CI.U Sig
1 Publication1    0.801 0.0685   11.69 52.30 0.000000000000000444   0.6635    0.938 ***
2 Publication2    0.371 0.1447    2.57  2.99 0.082973687325804235  -0.0899    0.833   *
---
Signif. codes: < .01 *** < .05 ** < .10 *
---
Note: If df < 4, do not trust the results> Wald_test(publication_effect, constraints = constrain_equal(1:2), vcov = "CR2")
 test Fstat df_num df_denom p_val sig
  HTZ   7.2      1     3.47 0.064   .
```

Figure B6. (a) Moderator analysis for publication type with an intercept and (b) moderator analysis for publication type without an intercept.
Note. Publication1 represents the category "Peer reviewed publications" and Publication2 represents the category "dissertation".



(a)

```
RVE: Correlated Effects Model with Small-Sample Corrections

Model: HedgesG ~ Control

Number of studies = 58
Number of outcomes = 112 (min = 1 , mean = 1.93 , median = 2 , max = 6 )
Rho = 0.8
I.sq = 89.0201
Tau.sq = 0.1942896

              Estimate StdErr t-value  dfs            P(|t|>) 95% CI.L 95% CI.U Sig
1 X.Intercept.    0.954 0.0844   11.30 38.1 0.0000000000000997    0.783    1.124 ***
2     Control2   -0.517 0.1284   -4.02 43.5 0.0002242352845960   -0.776   -0.258 ***
---
Signif. codes: < .01 *** < .05 ** < .10 *
---
Note: If df < 4, do not trust the results> Wald_test(control_effect, constraints = constrain_zero(2:2), vcov = "CR2")
 test Fstat df_num df_denom  p_val sig
  HTZ  16.2      1     43.5 <0.001 ***
```

(b)

```
          Estimate StdErr t-value  dfs            P(|t|>) 95% CI.L 95% CI.U Sig
1 Control1    0.954 0.0844   11.30 38.1 0.0000000000000997    0.783     1.12 ***
2 Control2    0.437 0.0982    4.45 22.8 0.0001889946811175    0.233     0.64 ***
---
Signif. codes: < .01 *** < .05 ** < .10 *
---
Note: If df < 4, do not trust the results> Wald_test(control_effect_noI, constraints = constrain_equal(1:2), vcov = "CR2")
 test Fstat df_num df_denom  p_val sig
  HTZ  16.2      1     43.5 <0.001 ***
```

Figure B7. (a) Moderator analysis for control group with an intercept and (b) moderator analysis for control group without an intercept.
Note. Control1 represents the category "non-clinical control" and Control2 represents the category "clinical control".

(a)

```
Model: HedgesG ~ Type

Number of studies = 58
Number of outcomes = 112 (min = 1 , mean = 1.93 , median = 2 , max = 6 )
Rho = 0.8
I.sq = 90.63074
Tau.sq = 0.2346392

              Estimate StdErr t-value  dfs         P(|t|>) 95% CI.L 95% CI.U Sig
1 X.Intercept.   0.7902 0.0864   9.148 39.5 0.0000000000266    0.616    0.965 ***
2        Type2  -0.0643 0.1272  -0.505 30.9 0.6168358033141   -0.324    0.195
---
Signif. codes: < .01 *** < .05 ** < .10 *
---
Note: If df < 4, do not trust the results> Wald_test(Type_effect, constraints = constrain_zero(2:2), vcov = "CR2")
 test Fstat df_num df_denom p_val sig
  HTZ 0.255      1     30.9 0.617
```

(b)

```
       Estimate StdErr t-value  dfs         P(|t|>) 95% CI.L 95% CI.U Sig
1 Type1    0.790 0.0864    9.15 39.5 0.0000000000266    0.616    0.965 ***
2 Type2    0.726 0.0934    7.77 15.9 0.0000008381641    0.528    0.924 ***
---
Signif. codes: < .01 *** < .05 ** < .10 *
---
Note: If df < 4, do not trust the results> Wald_test(Type_effect_noI, constraints = constrain_equal(1:2), vcov = "CR2")
 test Fstat df_num df_denom p_val sig
  HTZ 0.255      1     30.9 0.617
```

Figure B8. (a) Moderator analysis for method of categorising ADHD with an intercept and (b) moderator analysis for method of categorising ADHD without an intercept.
Note. Type1 represents the category "clinical" and Type2 represents the category "self-report".

(a)

```
RVE: Correlated Effects Model with Small-Sample Corrections

Model: HedgesG ~ Anxiety

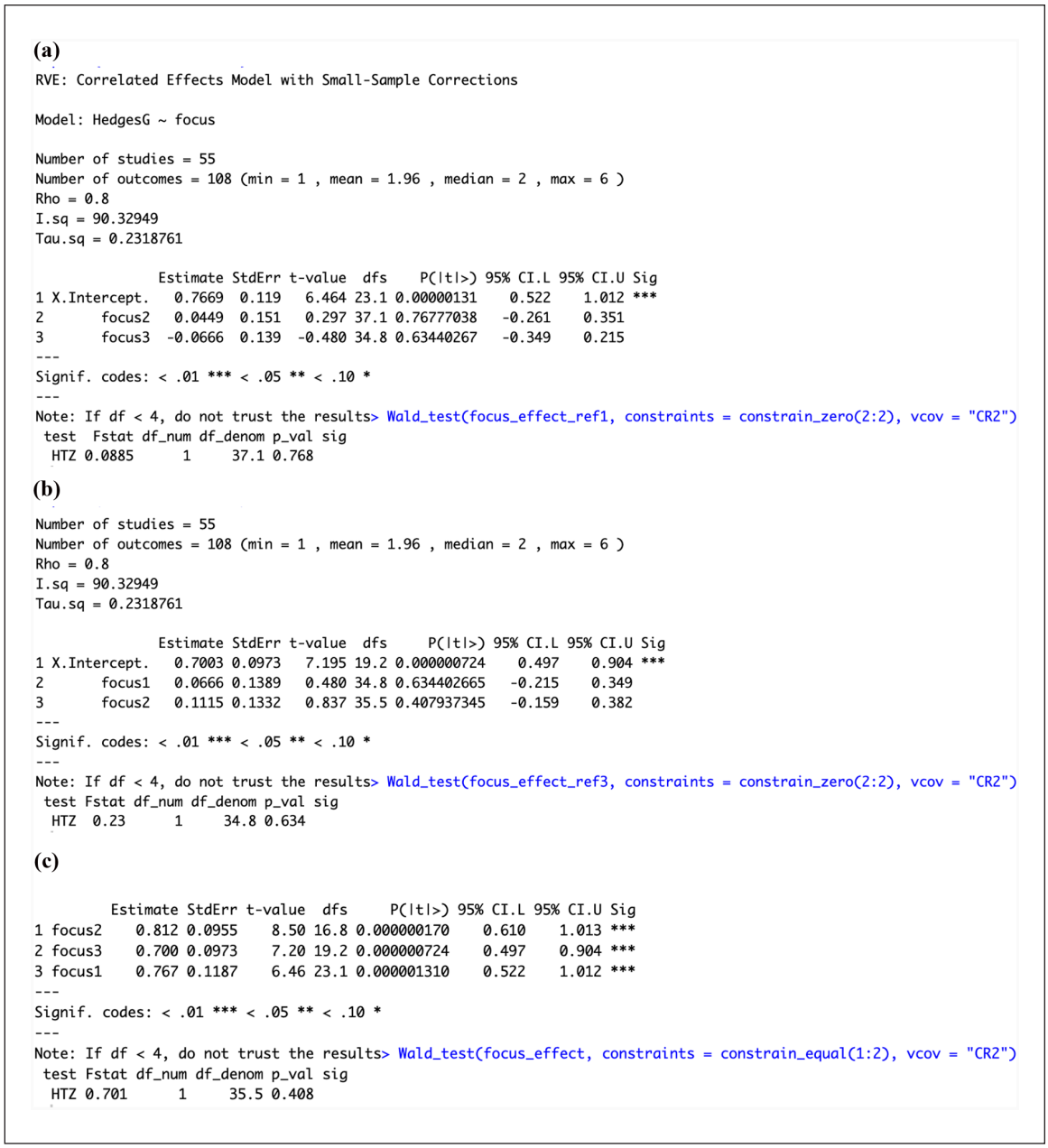
Number of studies = 58
Number of outcomes = 112 (min = 1 , mean = 1.93 , median = 2 , max = 6 )
Rho = 0.8
I.sq = 90.43419
Tau.sq = 0.2235848

              Estimate StdErr t-value  dfs              P(|t|>) 95% CI.L 95% CI.U Sig
1 X.Intercept.    0.816 0.0688   11.86 47.9 0.000000000000000888    0.677    0.954 ***
2     Anxiety2   -0.289 0.1877   -1.54 13.0 0.147641016612879650   -0.695    0.117
---
Signif. codes: < .01 *** < .05 ** < .10 *
---
Note: If df < 4, do not trust the results> Wald_test(Anxiety_effect, constraints = constrain_zero(2:2), vcov = "CR2")
 test Fstat df_num df_denom p_val sig
  HTZ  2.37      1       13 0.148
```

(b)

```
        Estimate StdErr t-value   dfs              P(|t|>) 95% CI.L 95% CI.U Sig
1 Anxiety1    0.816 0.0688   11.86 47.91 0.000000000000000888    0.677    0.954 ***
2 Anxiety2    0.527 0.1760    2.99  9.32 0.014597692349748392    0.130    0.923  **
---
Signif. codes: < .01 *** < .05 ** < .10 *
---
Note: If df < 4, do not trust the results> Wald_test(Anxiety_effect_noI, constraints = constrain_equal(1:2), vcov = "CR2")
 test Fstat df_num df_denom p_val sig
  HTZ  2.37      1       13 0.148
```

Figure B9. (a) Moderator analysis for anxiety measure type with an intercept and (b) moderator analysis for anxiety measure type without an intercept.
Note. Anxiety1 represents the category "general" and Anxiety2 represents the category "specific".



(a)

```
RVE: Correlated Effects Model with Small-Sample Corrections

Model: HedgesG ~ focus

Number of studies = 55
Number of outcomes = 108 (min = 1 , mean = 1.96 , median = 2 , max = 6 )
Rho = 0.8
I.sq = 90.32949
Tau.sq = 0.2318761

              Estimate StdErr t-value  dfs    P(|t|>) 95% CI.L 95% CI.U Sig
1 X.Intercept.   0.7669  0.119   6.464 23.1 0.00000131    0.522    1.012 ***
2       focus2   0.0449  0.151   0.297 37.1 0.76777038   -0.261    0.351
3       focus3  -0.0666  0.139  -0.480 34.8 0.63440267   -0.349    0.215
---
Signif. codes: < .01 *** < .05 ** < .10 *
---
Note: If df < 4, do not trust the results> Wald_test(focus_effect_ref1, constraints = constrain_zero(2:2), vcov = "CR2")
 test  Fstat df_num df_denom p_val sig
  HTZ 0.0885      1     37.1 0.768
```

(b)

```
Number of studies = 55
Number of outcomes = 108 (min = 1 , mean = 1.96 , median = 2 , max = 6 )
Rho = 0.8
I.sq = 90.32949
Tau.sq = 0.2318761

              Estimate StdErr t-value  dfs     P(|t|>) 95% CI.L 95% CI.U Sig
1 X.Intercept.   0.7003 0.0973   7.195 19.2 0.000000724    0.497    0.904 ***
2       focus1   0.0666 0.1389   0.480 34.8 0.634402665   -0.215    0.349
3       focus2   0.1115 0.1332   0.837 35.5 0.407937345   -0.159    0.382
---
Signif. codes: < .01 *** < .05 ** < .10 *
---
Note: If df < 4, do not trust the results> Wald_test(focus_effect_ref3, constraints = constrain_zero(2:2), vcov = "CR2")
 test Fstat df_num df_denom p_val sig
  HTZ  0.23      1     34.8 0.634
```

(c)

```
         Estimate StdErr t-value  dfs     P(|t|>) 95% CI.L 95% CI.U Sig
1 focus2    0.812 0.0955    8.50 16.8 0.000000170    0.610    1.013 ***
2 focus3    0.700 0.0973    7.20 19.2 0.000000724    0.497    0.904 ***
3 focus1    0.767 0.1187    6.46 23.1 0.000001310    0.522    1.012 ***
---
Signif. codes: < .01 *** < .05 ** < .10 *
---
Note: If df < 4, do not trust the results> Wald_test(focus_effect, constraints = constrain_equal(1:2), vcov = "CR2")
 test Fstat df_num df_denom p_val sig
  HTZ 0.701      1     35.5 0.408
```

Figure B10. (a) Moderator analysis for anxiety measure focus with an intercept set at category 1, (b) moderator analysis for anxiety measure focus with an intercept set at category 3, and (c) moderator analysis for anxiety measure focus without an intercept.

Note. Focus1 represents the category "state", Focus2 represents the category "prolonged-state", and Focus3 represents the category "trait"

Publication Bias Analysis

```
RVE: Correlated Effects Model with Small-Sample Corrections

Model: hedges ~ SEg

Number of studies = 58
Number of outcomes = 58 (min = 1 , mean = 1 , median = 1 , max = 1 )
Rho = 0.5
I.sq = 86.63898
Tau.sq = 0.1553126

               Estimate StdErr t-value  dfs P(|t|>) 95% CI.L 95% CI.U Sig
1 X.Intercept.    0.402  0.175    2.30 21.7  0.0316  0.03884    0.766  **
2          SEg    1.897  0.918    2.07 23.5  0.0500  0.00039    3.794  **
---
Signif. codes: < .01 *** < .05 ** < .10 *
---
Note: If df < 4, do not trust the results
```

Figure B11. Output of Robust Egger's regression – Precision Effect Estimate with Standard Error (PEESE).

```
RVE: Correlated Effects Model with Small-Sample Corrections

Model: hedges ~ Vg

Number of studies = 58
Number of outcomes = 58 (min = 1 , mean = 1 , median = 1 , max = 1 )
Rho = 0.5
I.sq = 86.45247
Tau.sq = 0.1514457

               Estimate StdErr t-value  dfs   P(|t|>) 95% CI.L 95% CI.U Sig
1 X.Intercept.    0.553  0.111    4.98 28.6 0.0000278    0.325     0.78 ***
2          Vg    5.025  2.428    2.07 15.8 0.0552927   -0.128    10.18   *
---
Signif. codes: < .01 *** < .05 ** < .10 *
---
Note: If df < 4, do not trust the results
```

Figure B12. Output of Robust Egger's regression – Precision-Effect Test (PET).

```
Estimated number of missing studies on the left side: 0 (SE = 4.5237)

Random-Effects Model (k = 58; tau^2 estimator: REML)

tau^2 (estimated amount of total heterogeneity): 0.2064 (SE = 0.0464)
tau (square root of estimated tau^2 value):      0.4543
I^2 (total heterogeneity / total variability):   89.72%
H^2 (total variability / sampling variability):  9.73

Test for Heterogeneity:
Q(df = 57) = 433.1080, p-val < .0001

Model Results:

estimate      se     zval    pval   ci.lb   ci.ub
  0.7685  0.0656  11.7177  <.0001  0.6399  0.8970  ***

---
Signif. codes:  0 '***' 0.001 '**' 0.01 '*' 0.05 '.' 0.1 ' ' 1
```

Figure B13. Output of trim-and-fill method.

Appendix C: Study ADHD Measures

Information regarding all the measures used to determine ADHD within studies included in the meta-analysis (Table C1).

Table C1. Details of ADHD Measures Used by Studies.

Acronym	Measure	References
ADHD-AFI	ADHD Adult Functioning Interview	Young, S., Toone, B., & Tyson, C. (2003). Comorbidity and psychosocial profile of adults with Attention Deficit Hyperactivity Disorder. *Personality and Individual Differences*, *35*(4), 743–755. https://doi.org/10.1016/S0191-8869(02)00267-2
ADHD BCA	The ADHD Behaviour Checklist for Adults	Murphy, K., & Barkley, R. A. (1996). Updated adult norms for the ADHD behavior checklist for adults. *ADHD Report*, *4*(4), 12–16.
ADHD-RS	ADHD Rating Scale	DuPaul, G. J., Power, T. J., Anastopoulos, A. D., & Reid, R. (1998). *ADHD Rating Scale—IV: Checklists, norms, and clinical inter pretation.* Guilford Press.
ASRS-V1.1	Adult ADHD Self-Report Scale-V1.1	Kessler, R. C., Adler, L., Ames, M., Demler, O., Faraone, S., Hiripi, E., Howes, M. J., Jin, R., Secnik, K., Spencer, T., Ustun, T. B., & Walters, E. E. (2005). The World Health Organization Adult ADHD Self-report Scale (ASRS): A short screening scale for use in the general population. *Psychological Medicine*, *35*(2), 245–256. https://doi.org/10.1017/S0033291704002892
ADDC	Attention Deficit Disorder Checklist	Kaplan, B. J., & Humphreys, R. (1995). Attention-deficit hyperactivity disorder checklist. Unpublished manuscript.
BAARS	Barkley's Adult ADHD Rating Scale	Barkley, R. A. (1998). *Attention-deficit hyperactivity disorder: A handbook for diagnosis and treatment* (2nd ed.). Guilford Press
BAARS-IV	Barkley Adult ADHD Rating Scales-IV	Barkley, R. A. (2011). *Barkley Adult ADHD Rating Scale-IV (BAARS-IV).* Guilford Press.
BCSS	Barkley's Current Symptom Scales	Barkley, R. A., & Murphy, K. R. (2006). *Attention-deficit hyperactivity disorder a clinical workbook (Vol. 3).* The Guilford Press.
BASC-2-C	Behaviour Assessment System for Children–Second Edition–College	Reynolds, C. R., & Kamphaus, R. W. (2004). *Behavior assessment system for children (2nd ed.) manual.* American Guidance Service.
CBPRS	Conner's Brief Parent Rating Scale	Conners, C. K. (1969). A teacher rating scale for use in drug studies with children. *The American Journal of Psychiatry*, *126*(6), 884–888. https://doi.org/10.1176/ajp.126.6.884
Conners 10	Conners 10-Symptom Abbreviated Parent Questionnaire	Conners, C. K. (1973). Rating scales for use in drug studies with children. *Psychopharmacology Bulletin*, *9*, 24–29.
CAADI	Conners' Adult ADHD Diagnostic Interview for Diagnostic and Statistical Manual Of Mental Disorders	Epstein, J. N., Johnson, D., & Conners, C. K. (2000). *Conners' adult ADHD diagnostic interview for DSM-IV*. Multi-Health Systems.
CAARS	Conners' Adult ADHD Rating Scale	Conners, C. K., Erhardt, D., & Sparrow, E. P. (2004). *Conners' Adult ADHD Rating Scales (CAARS): Technical manual.* MultiHealthSystems.
DIVA-2	Diagnostic Interview for ADHD in Adults	Kooij, J. J. S., & Francken, M. H. (2010). *Diagnostic interview for ADHD in adults 2.0 (DIVA 2.0)*. DIVA Foundation
DISC-IV	Diagnostic Interview Schedule for Children-IV–ADHD Module-Whole Life Version	Shaffer, D., Fisher, P., Lucas, C. P. et al. (2000). NIMH diagnostic interview schedule for children version IV (NIMH DISC-IV): Description, differences from previous versions, and reliability of some common diagnoses. *Journal of the American Academy of Child and Adolescent Psychiatry*, *39*, 28–38. https://doi.org/10.1097/00004583-200001000-00014
EISRB	Essen Interview-For-School-Days-Related Biography	Grabemann, M., Zimmermann, M., Strunz, L., Ebbert-Grabemann, M., Scherbaum, N., Kis, B., & Mette, C. (2017). Neue Wege in der Diagnostik der ADHS bei Erwachsenen [New ways of diagnosing ADHD in adults]. *Psychiatrische Praxis*, *44*(4), 221–227. https://doi.org/10.1055/s-0043-100714
AADHD	Korean Adult Attention Deficit/ Hyperactivity Disorder Scale	Kim, E. (2003). The validation of Korean adult ADHD scale (K-AADHDS). *Korean Journal of Clinical Psychology*, *22*, 897–911.
MINI	Mini-International Neuropsychiatric Interview	Sheehan, D. V., Lecrubier, Y., Sheehan, K. H., Amorim, P., Janavs, J., Weiller, E., Hergueta, T., Baker, R., & Dunbar, G. C. (1998). The Mini-International Neuropsychiatric Interview (M.I.N.I.): The development and validation of a structured diagnostic psychiatric interview for DSM-IV and ICD-10. *The Journal of Clinical Psychiatry*, *59*(Suppl. 20), 22–57.

(continued)

Table C1. (continued)

Acronym	Measure	References
K-SADS-PL	Schedule For Affective Disorders and Schizophrenia for School-Age Children-Present and Lifetime Version	Kaufman, J., Birmaher, B., Brent, D., Rao, U., Flynn, C., Moreci, P., Williamson, D., & Ryan, N. (1997). Schedule for Affective Disorders and Schizophrenia for School-Age Children-Present and Lifetime Version (K-SADS-PL): Initial reliability and validity data. *Journal of the American Academy of Child and Adolescent Psychiatry*, *36*(7), 980–988. https://doi.org/10.1097/00004583-199707000-00021
SDQ	Strengths And Difficulties Questionnaire	Goodman, R. (1997). The Strengths and Difficulties Questionnaire: A research note. *Journal of Child Psychology and Psychiatry, and Allied Disciplines*, *38*(5), 581–586. https://doi.org/10.1111/j.1469-7610.1997.tb01545.x
SCID-V	Structured Clinical Diagnostic Interview	First, M. (2015). Structured clinical interview for the DSM (SCID). In Cautin, R. L., & Lilienfeld, S. O. (Eds.) *Encyclopedia of clinical psychology*. John Wiley & Sons
NHIS	The National Hospital Interview Schedule	Robertson, M. M., & Eapen, V. (1996). The National Hospital Interview Schedule for the assessment of Gilles de la Tourette Syndrome. *International Journal of Methods in Psychiatric Research*, *6*(4), 203–226. https://doi.org/10.1002/(SICI)1234-988X(199612)6:4<203::AID-MPR165>3.3.CO;2-F
WURS	The Wender Utah Rating Scale	Ward, M. F., Wender, P. H., & Reimherr, F. W. (1993). The Wender Utah Rating Scale: An aid in the retrospective diagnosis of childhood attention deficit hyperactivity disorder. *The American Journal of Psychiatry*, *150*(6), 885–890. https://doi.org/10.1176/ajp.150.6.885
T-ADHD-SRS	Turgay's Adult Attention-Deficit/Hyperactivity Disorder DSM-IV-Based Diagnostic Screening and Rating Scale	Turgay, A. (1995). *Adult hyperactivity assessment scale based on DSM IV (unpublished scale).* Integrative Therapy Institute Toronto.
WRI	Wender-Reimherr-Interview	Retz-Junginger, P., Giesen, L., Philipp-Wiegmann, F., Rösler, M., & Retz, W. (2017). Der Wender-Reimherr-Selbstbeurteilungsfragebogen zur adulten ADHS: Deutsche Version [Wender-Reimherr self-report questionnaire on adult ADHD: German version]. *Der Nervenarzt*, *88*(7), 797–801. https://doi.org/10.1007/s00115-016-0110-4
YAQ	Young ADHD Questionnaire	Young, S. (2004). The YAQ-S and YAQ-I: The development of self and informant questionnaires reporting on current adult ADHD symptomatology, comorbid and associated problems. *Personality and Individual Differences*, *36*(5), 1211–1223. https://doi.org/10.1016/S0191-8869(03)00212-5

Note. References are included as stated in corresponding studies, for corresponding studies see Table 1 in text.

Appendix D: Study Anxiety Measures

Information regarding all the measures used to measure anxiety within studies included in the meta-analysis (Table D1).

Table D1. Details of Anxiety Measures Used by Studies.

Acronym	Measure	References
ASEBA	Achenbach System of Empirically Based Assessment–Anxiety subscale	Achenbach, T. M., & Rescorla, L. A. (2006). The Achenbach system of empirically based assessment. In R. P. Archer (Ed.), *Forensic uses of clinical assessment instruments.* Lawrence Erlbaum Associates.
ASAQ	Adult Separation Anxiety Questionnaire	Manicavasagar, V., Silove, D., Wagner, R., & Drobny, J. (2003). A self-report questionnaire for measuring separation anxiety in adulthood. *Comprehensive Psychiatry*, *44*(2), 146–153. https://doi.org/10.1053/comp.2003.50024
ASI	Anxiety Sensitivity Index	Peterson, R. A., & Reiss, S. (1992). *Anxiety sensitivity index manual.* International Diagnostic Systems.
BAI	Beck Anxiety Inventory	Beck, A. T., Epstein, N., Brown, G., & Steer, R. A. (1988). An inventory for measuring clinical anxiety: Psychometric properties. *Journal of Consulting and Clinical Psychology*, *56*(6), 893–897. https://doi.org/10.1037/0022-006X.56.6.893
BASC-2-C	Behavior Assessment System for Children–Second Edition–College–Anxiety subscale	Reynolds, C. R., & Kamphaus, R. W. (2004). *Behavior assessment system for children (2nd ed.) manual.* American Guidance Service.
DÄS-Zuf	DÄS-Zuf questionnaire – anxiety/depression subscale	Krieger, S. (2013). *Fragebogenstudie zur Untersuchung möglicher psychischer Erkrankungen und Sexueller Funktionsstörungen im Zusammenhang zu Lebens- und Selbstzufriedenheit bei Mukopolysaccharidose Patientinnen und Patienten* [*Questionnaire study to investigate possible mental illnesses and sexual dysfunctions in relation to life and self-satisfaction in mucopolysaccharidosis patients*] [Dissertation].
DASS	Depression and Anxiety Severity Scale	Lovibond, P. F., & Lovibond, S. H. (1995). The structure of negative emotional states: Comparison of the depression anxiety stress scales (DASS) with the beck depression and anxiety inventories. *Behaviour Research and Therapy*, *33*(3), 335–343. https://doi.org/10.1016/0005-7967(94)00075-u
DASS-21	Depression and Anxiety Severity Scale–Short form	Henry, J. D., & Crawford, J. R. (2005). The short-form version of the depression anxiety stress scale (DASS-21): Construct validity and normative data in a large non-clinical sample. *British Journal of Clinical Psychology*, *44*, 227–239. http://dx.doi.org/10.1348/014466505X29657
GAD 7	Generalised Anxiety Disorder Questionnaire	Spitzer, R. L., Kroenke, K., Williams, J. B., & Löwe, B. (2006). A brief measure for assessing generalized anxiety disorder: The GAD-7. *Archives of Internal Medicine*, *166*(10), 1092–1097. https://doi.org/10.1001/archinte.166.10.1092
HARS	Hamilton Anxiety Rating Scale	Hamilton, M. (1959). The assessment of anxiety states by rating. *British Journal of Medical Psychology*, *32*, 50–55. https://doi.org/10.1111/j.2044-8341.1959.tb00467.x
HADS	Hospital Anxiety and Depression Scale	Zigmond, A. S., & Snaith, R. P. (1983). The hospital anxiety and depression scale. *Acta Psychiatrica Scandinavica*, *67*(6), 361–370. https://doi.org/10.1111/j.1600-0447.1983.tb09716.x
LSAS	Liebowitz Social Anxiety Scale	Liebowitz, M. R. (1987). Social phobia. *Modern Problems of Pharmacopsychiatry*, *22*, 141–173. https://doi.org/10.1159/000414022
PSWQ	Penn State Worry Questionnaire	Meyer, T. J., Miller, M. L., Metzger, R. L., & Borkovec, T. D. (1990). Development and validation of the Penn State Worry Questionnaire. *Behaviour Research and Therapy*, *28*(6), 487–495. https://doi.org/10.1016/0005-7967(90)90135-6
SAS	Self-rating Anxiety Scale	Zung, W. W. (1971). A rating instrument for anxiety disorders. *Psychosomatics*, *12*(6), 371–379. https://doi.org/10.1016/S0033-3182(71)71479-0
SASI	Separation Anxiety Symptom Inventory	Silove, D., Manicavasagar, V., O'Connell, D., Blaszczynski, A., Wagner, R., & Henry, J. (1993). The development of the Separation Anxiety Symptom Inventory (SASI). *Australia and New Zealand Journal of Psychiatry*, *273*, 477–488.
STAI	State-Trait Anxiety Inventory	Spielberger, C. D., Gorsuch, R., Lushene, R., Vagg, P., & Jacobs, G. (1983). *Manual for the state-trait anxiety inventory.* Consulting Psychologists Press.
STAI-6	State-Trait Anxiety Inventory–short form	Marteau, T. M., & Bekker, H. (1992). The development of a six-item short-form of the state scale of the Spielberger State—Trait Anxiety Inventory (STAI). *British Journal of Clinical Psychology*, *31*(3), 301–306. https://doi.org/10.1111/j.2044-8260.1992.tb00997.x
SCL-90	Symptom Checklist 90 – Anxiety subscale	Derogatis, L. R., Lipman, R. S., & Covi, L. (1973). SCL-90: An outpatient psychiatric rating scale – Preliminary report. *Psychopharmacology Bulletin*, 9(1), 13–28.
TAI-5	Test Anxiety Inventory – 5	Taylor, J., & Deane, F. P. (2002). Development of a short form of the test anxiety inventory (TAI). *Journal of General Psychology*, *129*, 127–136. http://dx.doi.org/10.1080/00221300209603133
R-MARS	The Revised Mathematics Anxiety Rating Scale	Plake, B. S., & Parker, C. S. (1982). The development and validation of a revised version of the mathematics anxiety rating scale. *Educational and Psychological Measurement*, *42*, 551–557. http://dx.doi.org/10.3758/BF03198531
TTAS	Timed Test Anxiety Scale	Lewandowski, L., Gathje, R. A., Lovett, B. J., & Gordon, M. (2013). Test-taking skills in college students with and without ADHD. *Journal of Psychoeducational Assessment*, *31*(1), 41–52. https://doi.org/10.1177/0734282912446304

Note. References are included as stated in corresponding studies, for corresponding studies see Table 1 in text.